arXiv:2610.22740v2 [cs.AI] 22 Oct 2026

R^{2}Orch: Recursive Self-Improvement for Agent Orchestration via Self-Play within a Skill-Internalization Loop

Mingda Zhang[1] **Yun Huang**[1] **Qiang Huang**[2] **Yanjin Li**[3] **Rui Qian**[2]
Qichao Zhao[4] **Tiesunlong Shen**[5] **Na Zhao**[1] **Jian Wang**[1] **Xiaoying Tang**[1]

[1]The Chinese University of Hong Kong, Shenzhen, China [2]Fudan University, China [3]University of Illinois Urbana-Champaign, USA
[4]Tsinghua University, China [5]National University of Singapore, Singapore

mingdazhang@ieee.org qiianruii@gmail.com

Abstract

In recent years, agentic orchestration has been widely applied to solve complex human tasks. However, existing orchestration learning still faces key challenges, including *hand-designed curricula*, *advantage collapse* when a task's rollouts all reach the same outcome, and *unreliable evaluation signals* that judge each skill by one noisy score. To address these challenges, we propose R^{2}Orch, a new paradigm of *Recursive Self-Play Orchestration*: an inner loop of self-play that improves the orchestrator on self-chosen tasks, and an outer loop that improves how the inner loop improves. To support this paradigm, we introduce *Recursive Learnability*, a training objective that rewards a Proposer for choosing tasks whose rollouts keep reaching different outcomes as skills are withdrawn. Built on this objective, we further propose *Skill Withdrawal*, in which the outer loop withdraws each skill once a controlled evaluation within self-play shows that the orchestrator performs as well without it. Moreover, the two loops form one cycle: failures are distilled into skills, skills into parameters, and withdrawals lead the Proposer to new failures, so the orchestrator moves toward zero-skill deployment. Experimental results on twelve datasets show that R^{2}Orch consistently outperforms baselines across question answering, math, code, and interactive tasks. Our code is available at github.com/beita6969/R2Orch.

1 Introduction

In recent years, a variety of powerful agentic systems have been applied to solve a wide range of human problems (Zhang et al., 2025; Li et al., 2026; Zhang et al., 2026c), gradually moving beyond hand-written agent pipelines toward orchestrators that learn to compose agent teams.

In this process, the agent graph has become a key bridge from task goals to executable collaboration: by giving each node a role and each edge a hand-off, orchestrators solve tasks with improved



Figure 1: A recursion needs two channels: the outer loop sets the inner loop's objective, and the inner loop, Proposer–Director self-play on a randomized rollout group, returns the evidence for withdrawing skills.

controllability, compositionality, and reusability (Zhang et al., 2025, 2026c), as shown in Figure 1. However, in practice, orchestrators still learn from tasks people collect and lean on in-prompt skills (Li et al., 2026; Zhang et al., 2026d), making them costly to scale and hard to improve on their own.

To address these issues, three main lines of work have emerged, as shown in Figure 2. First, *learned orchestration* optimizes workflows and planning from task feedback (Zhang et al., 2025; Li et al., 2026; Zhang et al., 2026c), taking the task pool as given. Second, *self-improving agents* refine their own memory, harness code, answers, or tasks through repeated loops (Zhu et al., 2026; Srikanth et al., 2026; Zhang et al., 2026b; Lu et al., 2026a; Fu et al., 2026). Third, *skill scaffolds* hand procedural knowledge to a policy (Zhang et al., 2026d; Shi et al., 2026; Lu et al., 2026b; Shen et al., 2026), in context at inference or as training-time support.

However, these methods still face three challenges (Figure 2a–c). **(i) Hand-designed curricula.** Learned orchestration and skill internalization practice on task pools that people collect in advance

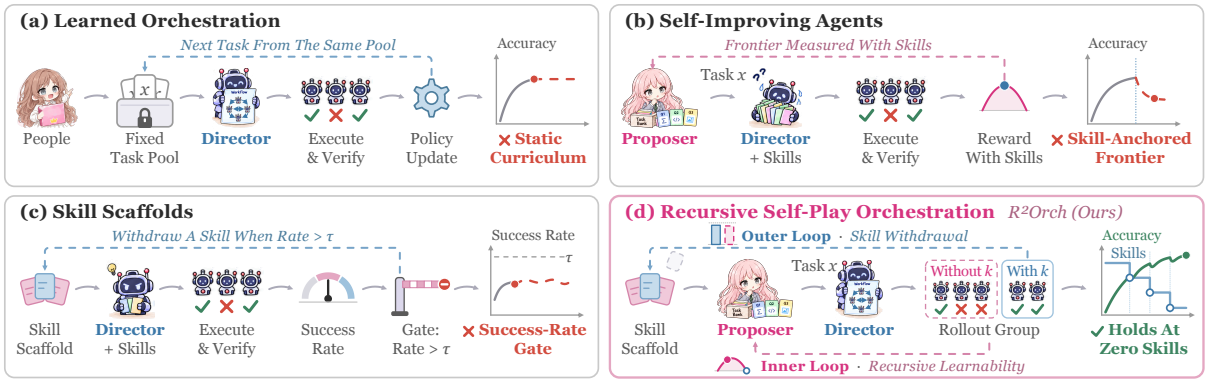


Figure 2: Paradigm comparison. (a) Learned orchestration trains on a hand-designed *static curriculum*. (b) Self-improving agents measure the frontier with skills present, a *skill-anchored frontier* where advantages collapse once skills leave. (c) Skill scaffolds withdraw guidance through *success-rate gates*, unreliable evaluation signals pinned by self-play. (d) R^{2}Orch nests two loops joined by downward (objective) and upward (evidence) channels.

(Zhang et al., 2026c; Shi et al., 2026), and self-play chooses its own tasks while the solver keeps retrieving its skills (Fu et al., 2026), so choosing tasks and withdrawing skills do not inform each other. **(ii) Advantage collapse.** Group-based RL such as GRPO learns from differences among a task's rollouts, which vanish when they all reach the same outcome (He et al., 2026); self-play keeps tasks at the solver's frontier with skills in context (Fu et al., 2026), so withdrawing a skill pushes the tasks it supported toward all-wrong rollouts, and advantage collapse returns. **(iii) Unreliable evaluation signals.** Training-time scaffolds fade guidance on a success rate mixing task choice with the skill's effect (Shi et al., 2026), and skill libraries keep or retire skills on one noisy score each, so they drift until injecting a skill scores worse than injecting nothing (Zhang et al., 2026e; Zhang et al.).

To address these challenges, we propose **R^{2}Orch** (Fig. 2d), a new paradigm of *Recursive Self-Play Orchestration*, in which an inner loop of self-play improves the orchestrator, a Director, on tasks a Proposer chooses, and an outer loop improves how the inner loop improves. Two channels join the loops, one carrying the objective down and one the evidence up. To support this paradigm, we introduce *Recursive Learnability*, the downward channel, which rewards the Proposer for choosing tasks whose rollouts keep reaching different outcomes as skills are withdrawn, scoring each task with the candidate skill withheld. Built on this objective, we further propose *Skill Withdrawal*, the upward channel, in which a random part of each rollout group hides one skill, turning self-play into a controlled evaluation whose Bayesian posterior over the skill's remaining benefit decides when it can leave. The two loops thus close into one cycle: failures are distilled into skills, skills into parameters, and withdrawals lead the Proposer to new failures, so the Director improves on its own and moves toward zero-skill deployment.

We evaluate R^{2}Orch on twelve benchmarks in question answering, math, code, and interactive tasks. Results show that R^{2}Orch outperforms all baselines on every benchmark, improves six other Worker backbones without retraining, and keeps its accuracy when tested with zero skills. Together, these results mark a step toward the recursive self-improvement of agent orchestrators.

2 Related Work

Learned Orchestration and Self-Play. Prior work searches agent workflows (Zhang et al., 2025), learns graph editing (Zhang et al., 2026c), or trains the planner of an agent system (Li et al., 2026), over tasks that people collect in advance. Self-play lets a Proposer pose tasks to a solver, rewarding regret (Dennis et al., 2020), solver uncertainty (Huang et al., 2026), or learning progress while distilling failures into skills (Fu et al., 2026). Group-based RL (Shao et al., 2024; Liu et al., 2025) gives no advantage to a task whose rollouts all reach the same outcome (He et al., 2026), so curricula favor learnable tasks (Rutherford et al., 2024) or filter the rest out (Yu et al., 2026). R^{2}Orch scores tasks with the candidate skill withheld, so its Proposer picks tasks whose rollouts keep reaching different outcomes instead of filtering collapsed ones.

Skill Evolution and Self-Improvement. Skills enter the prompt at inference (Zhang et al., 2026d) or support training and are withdrawn on a success-

rate threshold (Shi et al., 2026), a preset budget (Lu et al., 2026b), or leave-one-skill-out validation (Shen et al., 2026). Rollouts with and without a skill score it (Tu et al., 2026; Lin et al., 2026), paired tests accept self-modifications (Shawn, 2026), and one noisy score per skill lets a library drift (Zhang et al., 2026e). Recursive self-improvement systems revise their own memory, code, or answers (Zhu et al., 2026; Srikanth et al., 2026; Lu et al., 2026a; Zhang et al., 2026b; Wang et al., 2026) or adopt a self-modification once it is shown to help (Schmidhuber, 2007; Zhang et al., 2026a). R^{2}Orch nests self-play in a skill-withdrawal loop and decides each withdrawal from the controlled evaluation within every randomized rollout group, at no extra rollouts.

3 Preliminaries

Definition 1: Agent Graph. An agent graph is a directed graph $\mathcal{G} = (V, E, \kappa, o)$, where each node $v \in V$ is an agent with a role prompt and a model route, each edge $e \in E$ is a directed or bidirectional hand-off between two agents, κ assigns every agent a layer, and $o \in V$ is the output agent whose answer is returned (our role-prompted nodes extend the operator nodes of Zhang et al., 2026c).

Definition 2: Director Trajectory. A complete sequence of T atomic edits from an empty agent graph to FINISH defines a trajectory, in which every edit is validated and applied before the next one and its feedback returns to the Director:

$$\begin{aligned} \xi = (a_1, \ldots, a_T) \Rightarrow g, \\ \mathcal{G}_t = \textsc{Apply}(\mathcal{G}_{t-1}, a_t), \;\; \mathcal{G}_0 = \varnothing, \;\; g = \mathcal{G}_T, \end{aligned} \tag{1}$$

where a_t is one of ADDAGENT, SETPROMPT, SETMODEL, SETLAYER, CONSIDERRELATION, DELETEAGENT, SETOUTPUT, and FINISH, and g is the final agent graph that frozen Workers execute.

Problem Statement. Given an evaluation distribution ρ, a frozen Worker pool, a verifier r, and a skill library, we seek a Director π_θ that builds a successful agent graph with no skill in its context, trained on tasks chosen by a Proposer ρ_ϕ:

$$\max_\theta \; \mathbb{E}_{x\sim\rho}\, \mathbb{E}_{g\sim\pi_\theta(\cdot|x,\varnothing)} \big[r(g,x) \big], \tag{2}$$

where $r(g,x) \in \{0,1\}$ is the verifier's correctness after the Workers execute g on x, and the empty set marks zero-skill deployment. Skills scaffold training, and the Proposer chooses tasks whose rollouts reach different outcomes. Equation (2) is the deployed target; Section 4 shows how a shrinking scaffold and a learned curriculum approach it.

4 Method

As illustrated in Figure 3, this section introduces the R^{2}Orch framework, including Recursive Self-Play Orchestration (Section 4.1), Recursive Learnability (Section 4.2), and Skill Withdrawal (Section 4.3), all built on one randomized rollout group.

4.1 Recursive Self-Play Orchestration

Recursive Self-Play Orchestration nests two self-improvement loops (Figure 3, top). In the inner loop, self-play improves the Director on tasks that the Proposer chooses; the outer loop improves how the inner loop improves, by withdrawing the Director's skills as each stops helping.

Two time scales. Following two-time-scale stochastic approximation (Borkar, 1997), both policies take proximal best responses on the fast time scale, each a clipped step (Schulman et al., 2017):

$$\begin{aligned} \theta_{t+1} &\approx \arg\max_\theta \mathbb{E}\, v_K(x;\theta) - \tfrac{1}{\beta}\,\mathrm{KL}(\pi_\theta \| \pi_{\theta_t}), \\ \phi_{t+1} &\approx \arg\max_\phi \mathbb{E}\, R_P(x) - \tau\,\mathrm{KL}(\rho_\phi \| \rho_0). \end{aligned} \tag{3}$$

Here the Director, trained on the Proposer's picks with $K \sim q_x$, is scored by the value $v_K(x;\theta) = \mathbb{E}_{\xi\sim\pi_\theta(\cdot|x,K)}[u(\xi,x)]$ of the context it reads under the reward u (correctness less a Worker cost), and the Proposer by the reward R_P of Section 4.2. On the slow scale, the outer loop updates the scaffold from the same rollout groups, and the scaffold in turn sets the randomized context of every group:

$$\begin{aligned} \mathcal{S}_{t+1} &= \Phi\big(\mathcal{S}_t;\, \{\Pi_k\}_{k\in\mathcal{S}_t},\, \mathcal{F}_t\big), \\ q_x &= \tfrac{n_h}{G}\,\delta_{K_x\setminus k_x} + \tfrac{n_v}{G}\,\delta_{K_x}, \end{aligned} \tag{4}$$

where $K_x \subseteq \mathcal{S}_t$ are the skills retrieved for x, $k_x \in K_x$ is the skill under test, Π_k the posterior over k's benefit, $\mathcal{F}_t$ the groups in which both parts fail, and Φ the decisions of Section 4.3; of the $G = n_h + n_v$ rollouts, rollout i reads K_i, which is $K_x \setminus k_x$ in the hidden part $\mathcal{H}_x$ ($n_h > n_v$) and K_x in the visible part $\mathcal{V}_x$. Because $\mathcal{S}_t$ constrains both q_x and the context on which R_P is computed, each outer step rewrites the objectives of the inner loop.

Director update. The randomized context is also how the inner loop prepares each withdrawal: because q_x puts most of its mass on $K_x \setminus k_x$, most of each update already trains the Director without the candidate skill. The Director uses the group-relative estimator of GRPO (Shao et al., 2024; Liu et al., 2025) with this adaptation; with centered advantages, it is a leave-one-out gradient (Ahmadian

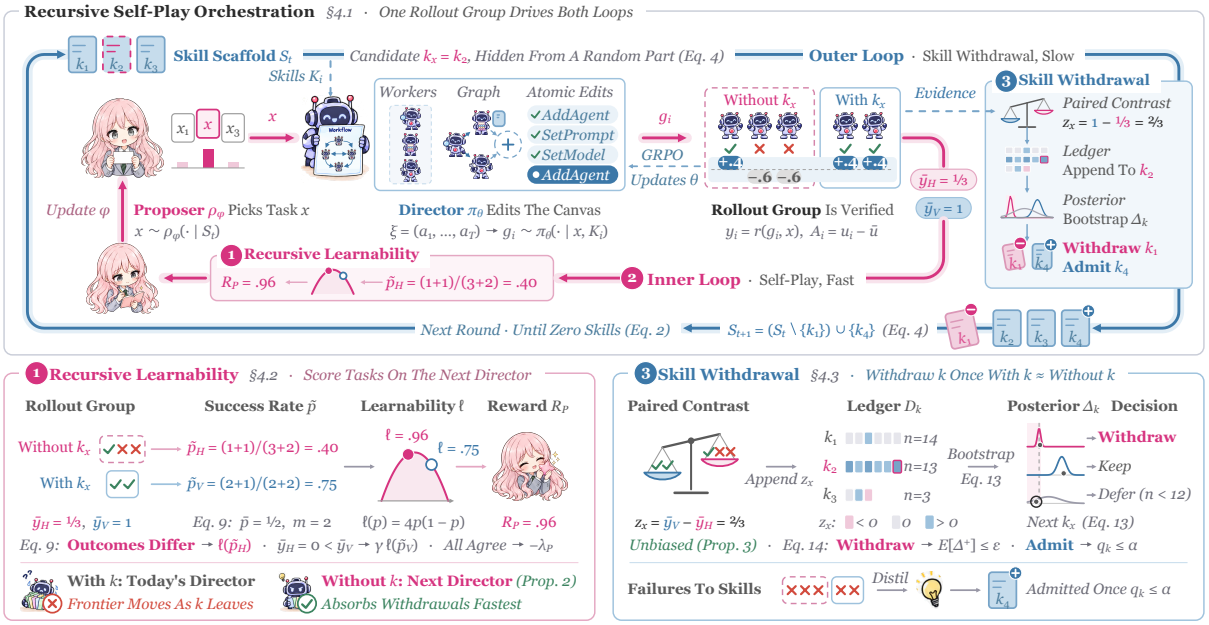


Figure 3: Overview of R^{2}Orch. Top: Proposer–Director self-play (inner loop) and skill withdrawal (outer loop) share one randomized rollout group. Bottom left: Recursive Learnability scores a task with the candidate skill hidden. Bottom right: Skill Withdrawal withdraws or admits each skill by the posterior of its paired contrast.

et al., 2024) of the first objective in Eq. (3):

$$\nabla_\theta \mathbb{E}_{K\sim q_x}\big[v_K(x;\theta)\big] = \mathbb{E}\Big[\tfrac{1}{G-1}\textstyle\sum_{i=1}^{G}(u_i-\bar{u}) \cdot \nabla_\theta \log \pi_\theta(\xi_i \mid x, K_i)\Big], \tag{5}$$

where ξ_i is the edit trajectory of rollout i (Definition 2) and $u_i - \bar{u} = (1 - \frac{1}{G})(u_i - \bar{u}_{-i})$ is the leave-one-out baseline of rollout i.

What each loop contributes. To see how the two loops share the work, let $J_{\mathcal{S}}(\theta) = \mathbb{E}_{x\sim\rho}[p_{K_x}(x)]$ be the success rate under scaffold $\mathcal{S}$ on ρ, with $p_K(x)$ the success probability under context K. Over the rounds it telescopes into inner gains $\Gamma_t = J_{\mathcal{S}_t}(\theta_{t+1}) - J_{\mathcal{S}_t}(\theta_t)$ and outer costs $C_t = J_{\mathcal{S}_t}(\theta_{t+1}) - J_{\mathcal{S}_{t+1}}(\theta_{t+1})$:

$$J_{\mathcal{S}_T}(\theta_T) = J_{\mathcal{S}_0}(\theta_0) + \textstyle\sum_{t<T}\big(\Gamma_t - C_t\big). \tag{6}$$

Here Γ_t is what self-play adds on a fixed scaffold, and C_t is what the scaffold change takes away: for one withdrawn skill k, its benefit $p_{K_x} - p_{K_x\setminus k}$ weighted by how often k is retrieved. With $\mathcal{S}_T = \varnothing$ the left side is the zero-skill objective, and the other two components each serve one term: Recursive Learnability drives Γ_t (Section 4.2), and Skill Withdrawal limits the withdrawal cost in each C_t (Section 4.3).

Proposition 1. *Zero-skill success is the starting success plus inner gains less outer costs, and self-play tracks each scaffold's equilibrium. Proof.* Appendix A, via Lemmas A.5 and A.6, following the two-time-scale analysis of Borkar (1997).

4.2 Recursive Learnability

Recursive Learnability is the downward channel, through which the outer loop sets the inner loop's objective: it rewards the Proposer for tasks whose rollouts keep reaching different outcomes with the candidate skill withheld (Figure 3, bottom left).

Tilt and its log-partition. How much a task can raise its success follows from the proximal step itself. Following relative-entropy policy search (Peters et al., 2010) and KL-regularized RL as inference (Korbak et al., 2022), a step of size β tilts the Director toward success, and its log-partition is the outcome's cumulant generating function:

$$\pi_\beta(g \mid x, K) = \pi_\theta(g \mid x, K)\, e^{\beta r(g,x)}/Z_K, \qquad \log Z_K = \beta p_K + \tfrac{\beta^2}{2} p_K(1-p_K) + O(\beta^3). \tag{7}$$

Here $Z_K(x;\beta) = 1 + p_K(e^\beta - 1)$ with $p_K = p_K(x)$, so the slope and curvature of $\log Z_K$ at $\beta = 0$ are the success probability and the outcome variance. A step gains $\beta\, p_K(1-p_K)$ in success at a KL cost of $\beta^2 p_K(1-p_K)/2$ to leading order, so within a KL budget the gain grows with the curvature.

Recursive target. On the current Director, this curvature is the learnability of the task (Rutherford et al., 2024). The progress that counts, however, is what survives withdrawal (Eqs. (5) and (6)), so

Recursive Learnability takes the curvature on the context with the candidate skill withheld:

$$\begin{aligned}\ell_{\mathrm{rec}}(x) &:= 4\,\partial_\beta^2 \log Z_{K_x\setminus k_x}\big|_{\beta=0} \\ &= 4\,p^-(1-p^-) = \tfrac{4}{G-1}\,\mathbb{E}\big[\textstyle\sum_i (y_i^- - \bar{y}^-)^2\big],\end{aligned} \tag{8}$$

a learnability proportional to the expected advantage energy of the group, where $p^- = p_{K_x\setminus k_x}(x)$, y_i^- are the outcomes of G rollouts that read $K_x \setminus k_x$ with mean $\bar{y}^-$, and $\ell(p) = 4p(1-p)$. The target vanishes exactly when $p^- \in \{0,1\}$, where every group collapses, and peaks at one half; since each withdrawal removes one more skill from $K_x \setminus k_x$, it follows the scaffold down to the zero-skill Director.

Proposer reward. The hidden part of each group already reads $K_x \setminus k_x$, so it measures p^- at no extra cost. The reward is an empirical-Bayes plug-in estimate of ℓ_{rec} on this part (Gelman et al., 1995), discounted by $\gamma < 1$ when only the visible part progresses and penalized by λ_P on collapse:

$$\begin{aligned}R_P(x) &= (1-c_x)\,\gamma^{h_x}\,\ell(\tilde{p}_x) - \lambda_P\,c_x, \\ \tilde{p}_x &= \mathbb{E}\big[p \mid y_{\mathcal{A}_x}\big] = \frac{\sum_{i\in\mathcal{A}_x} y_i + m\,\bar{p}}{|\mathcal{A}_x| + m}.\end{aligned} \tag{9}$$

Here $c_x = \mathbf{1}[\bar{y} \in \{0,1\}]$ flags collapse, $h_x = \mathbf{1}[\bar{y}_{\mathcal{H}} = 0 < \bar{y}_{\mathcal{V}}]$ switches $\mathcal{A}_x$ from $\mathcal{H}_x$ to $\mathcal{V}_x$, and the Beta prior shrinks toward the dataset's running hidden-part success rate $\bar{p}$ with strength m.

Proposer update. These rewards become a curriculum through the Proposer. As a KL-regularized environment designer (Dennis et al., 2020), it maximizes $J_P(\rho) = \mathbb{E}_{x\sim\rho}[R_P(x)] - \tau\,\mathrm{KL}(\rho\|\rho_0)$ over the candidate list C_t, whose optimum is the closed-form Gibbs curriculum (Lemma A.9):

$$\rho^\star(x) = \frac{\rho_0(x)\,e^{\bar{R}_P(x)/\tau}}{\sum_{x'\in C_t}\rho_0(x')\,e^{\bar{R}_P(x')/\tau}}. \tag{10}$$

With $\bar{R}_P(x)$ the expected reward of task x, the curriculum tilts the base model ρ_0 toward tasks of high Recursive Learnability. A pairwise update, with each pick as the other's baseline (Ahmadian et al., 2024), moves the Proposer toward these tasks.

Proposition 2. *Under a small trust region, a task's success gain grows with its Recursive Learnability. Proof.* Appendix A, via Lemmas A.1–A.3 on tilt and energy (graph-determined outcomes).

4.3 Skill Withdrawal

Skill Withdrawal is the upward channel, through which the inner loop supplies the outer loop's evidence: it keeps each C_t small by withdrawing a skill once the same rollout groups, read as a controlled evaluation, show that the Director performs as well without it (Figure 3, bottom right).

Estimand and estimator. Because the hidden part is drawn at random, each rollout group is a randomized block (Rubin, 1974; Imbens and Rubin, 2015), and the difference of its part means is an unbiased (Neyman) estimator of the skill's effect:

$$\begin{aligned}z_x &= \tfrac{1}{n_v}\textstyle\sum_{i\in\mathcal{V}_x} y_i - \tfrac{1}{n_h}\sum_{i\in\mathcal{H}_x} y_i, \\ \mathbb{E}[z_x \mid x, k_x{=}k] &= p_{K_x} - p_{K_x\setminus k} =: \Delta_k(x),\end{aligned} \tag{11}$$

where $k = k_x$. Averaging over the tasks that test k gives its benefit $\Delta_k = \mathbb{E}[\Delta_k(x) \mid k_x = k]$, which estimates its cost per retrieval in C_t (Lemma A.5). The success rate mixes it with the curriculum:

$$\mathbb{E}[\bar{y} \mid x, k_x = k] = \underbrace{p_{K_x\setminus k}(x)}_{\text{curriculum}} + \underbrace{\tfrac{n_v}{G}\,\Delta_k(x)}_{\text{skill}}. \tag{12}$$

Recursive Learnability pulls the curriculum term toward one half, so a threshold on $\bar{y}$ reads the curriculum rather than the skill, while z_x removes it.

Posterior and decisions. The contrasts accumulate into a Bayesian-bootstrap posterior per skill (Rubin, 1981), and a Thompson draw tests first the skill closest to withdrawal (William et al., 1933):

$$\begin{aligned}\Pi_k &= \mathrm{Law}\big(\textstyle\sum_{x\in\mathcal{D}_k} w_x z_x\big), \quad w \sim \mathrm{Dir}(\mathbf{1}_{n_k}), \\ k_x &= \arg\min_{k\in K_x}\ \tilde{\Delta}_k, \qquad \tilde{\Delta}_k \sim \Pi_k^0,\end{aligned} \tag{13}$$

where $\mathcal{D}_k$ holds the n_k records of skill k, and Π_k^0 adds one prior record so that untested skills are also drawn. On this posterior, withdrawal and admission are Bayes decisions (Berger, 1985) under loss Δ^+ for withdrawing, ϵ for keeping, and an asymmetric zero-one loss for admitting:

$$\begin{aligned}\mathcal{W}_t &= \big\{k :\ n_k \ge n_{\min},\ \mathbb{E}_{\Delta\sim\Pi_k}[\Delta^+] \le \epsilon\big\}, \\ \mathcal{N}_t &= \big\{k :\ n_k \ge n_{\min},\ q_k \le \alpha\big\},\end{aligned} \tag{14}$$

with $\Delta^+ = \max(\Delta, 0)$, $q_k = \Pi_k(\Delta \le 0)$, $n_{\min}$ the minimum record, and α the admission risk, so each withdrawal costs at most ϵ in posterior expectation on the tasks on which its skill was tested.

Failures to skills. The same posterior closes the outer loop: the least helpful skill is revised first, and new skills are distilled from groups where both parts fail (Appendix C), so failures exposed by withdrawals become the next skills to internalize.

Proposition 3. *Random hiding makes the paired contrast unbiased for a skill's benefit. Proof.* Appendix A, via randomization (Rubin, 1974).

		Question answering					Math	Code	Interactive	Average		
(a) In-distribution		HotpotQA		NQ-Open		MedQA	AIME 2026	MBPP+	ALFWorld	IID		
Type	**Method**	EM	F1	EM	F1	Acc.	Acc.	Pass@1	SR	EM	F1	Acc.
Backbone	Qwen3.5-9B	57.81 ±0.00	73.75 ±0.22	23.59 ±0.86	32.89 ±1.39	70.94 ±0.86	48.67 ±2.98	78.59 ±0.70	47.19 ±0.70	40.70 ±0.43	53.32 ±0.70	61.35 ±0.84
	DeepSeek-V4-Flash	71.25 ±1.28	82.50 ±0.85	38.44 ±0.86	49.77 ±0.89	84.38 ±0.00	49.33 ±3.65	84.69 ±0.70	60.16 ±1.10	54.84 ±0.71	66.14 ±0.62	69.64 ±1.07
Training	SFT	63.28 ±1.10	77.58 ±0.89	25.78 ±0.00	37.50 ±0.19	73.59 ±0.86	34.67 ±1.83	78.28 ±0.86	40.94 ±0.43	44.53 ±0.55	57.54 ±0.46	56.87 ±0.59
	GRPO	69.22 ±0.43	81.33 ±0.43	24.53 ±0.70	32.27 ±1.41	77.03 ±0.70	34.00 ±4.94	82.19 ±0.65	50.94 ±1.02	46.88 ±0.48	56.80 ±0.74	61.04 ±1.60
Search	AFlow	87.66 ±0.65	88.83 ±0.67	76.56 ±0.78	81.09 ±0.63	89.22 ±0.35	52.00 ±2.98	85.31 ±0.86	73.44 ±0.00	82.11 ±0.33	84.96 ±0.46	74.99 ±0.92
Agent RL	AgentFlow	86.25 ±0.89	88.44 ±1.98	81.41 ±0.86	85.78 ±0.98	88.44 ±1.69	61.33 ±1.83	85.63 ±0.43	76.09 ±0.70	83.83 ±0.59	87.11 ±1.10	77.87 ±0.35
	FlowSteer	87.97 ±0.70	88.91 ±1.49	83.91 ±0.43	86.88 ±0.49	90.94 ±0.70	67.33 ±3.65	88.91 ±1.02	82.19 ±0.86	85.94 ±0.28	87.90 ±0.78	82.34 ±1.00
Skill int.	SKILL0	87.34 ±0.86	88.52 ±1.49	82.97 ±0.35	86.48 ±0.87	89.53 ±0.70	64.67 ±6.91	87.97 ±1.18	83.44 ±0.35	85.16 ±0.39	87.50 ±0.86	81.40 ±1.65
	PATS	89.06 ±1.46	91.25 ±1.59	84.22 ±1.16	87.19 ±1.21	91.41 ±1.46	74.00 ±3.65	90.63 ±0.55	87.97 ±0.70	86.64 ±0.80	89.22 ±1.00	86.00 ±0.95
Self-play	SESA	88.13 ±1.40	90.39 ±1.23	83.28 ±0.43	86.95 ±0.67	88.75 ±0.70	68.00 ±5.06	88.44 ±0.86	81.09 ±0.65	85.70 ±0.59	88.67 ±0.70	81.57 ±1.31
	MetaSkill	87.03 ±1.42	88.75 ±0.64	82.50 ±0.70	85.63 ±0.53	88.44 ±0.86	55.33 ±2.98	86.25 ±1.52	79.06 ±0.86	84.77 ±0.83	87.19 ±0.42	77.27 ±1.21
Ours	**R^{2}Orch**	**91.09** ±0.43	**92.34** ±0.54	**86.09** ±0.65	**89.30** ±0.76	**94.06** ±0.70	**81.33** ±2.98	**93.13** ±0.35	**90.63** ±0.96	**88.59** ±0.43	**90.82** ±0.47	**89.79** ±1.01
(b) Out-of-distribution		TriviaQA		MuSiQue		GPQA	MATH-Hard	SWE-Bench	WebShop	OOD		
Type	**Method**	EM	F1	EM	F1	Acc.	Acc.	Resolved	SR	EM	F1	Acc.
Backbone	Qwen3.5-9B	44.53 ±0.00	53.44 ±0.06	39.22 ±0.65	48.98 ±0.92	61.41 ±0.70	89.38 ±0.43	15.78 ±0.86	32.81 ±0.78	41.88 ±0.33	51.21 ±0.46	49.84 ±0.29
	DeepSeek-V4-Flash	70.31 ±1.66	80.94 ±1.91	50.31 ±0.43	58.59 ±0.69	74.22 ±1.66	92.50 ±1.31	35.78 ±0.86	65.31 ±1.05	60.31 ±0.76	69.77 ±1.02	66.95 ±0.58
Training	SFT	46.41 ±0.70	58.91 ±0.83	38.75 ±0.70	50.78 ±0.93	68.91 ±0.65	87.97 ±0.70	14.69 ±0.65	32.97 ±1.40	42.58 ±0.28	54.85 ±0.62	51.13 ±0.51
	GRPO	47.66 ±0.55	58.67 ±0.68	40.94 ±1.52	50.08 ±1.67	67.03 ±0.86	87.81 ±0.70	17.50 ±0.43	35.94 ±0.78	44.30 ±0.90	54.38 ±0.90	52.07 ±0.30
Search	AFlow	90.31 ±0.70	92.19 ±1.04	77.19 ±1.50	82.89 ±1.75	75.78 ±0.78	91.41 ±0.00	28.28 ±0.86	59.38 ±0.78	83.75 ±0.90	87.54 ±1.02	63.71 ±0.49
Agent RL	AgentFlow	88.91 ±1.28	90.78 ±1.65	79.06 ±1.50	84.06 ±1.72	79.53 ±0.35	93.13 ±0.65	39.38 ±1.31	75.94 ±0.65	83.98 ±0.92	87.42 ±1.19	71.99 ±0.49
	FlowSteer	92.50 ±1.62	93.83 ±1.89	79.69 ±1.10	84.77 ±1.43	83.91 ±0.43	93.44 ±0.43	37.97 ±0.70	78.75 ±1.40	86.09 ±0.98	89.30 ±1.19	73.52 ±0.26
Skill int.	SKILL0	91.25 ±0.65	93.13 ±0.74	79.22 ±0.70	82.97 ±0.88	81.25 ±0.78	92.50 ±1.05	36.88 ±0.86	82.81 ±1.56	85.23 ±0.64	88.05 ±0.57	73.36 ±0.47
	PATS	93.75 ±0.00	94.53 ±0.17	82.97 ±0.65	86.64 ±0.73	82.97 ±0.65	93.44 ±0.70	40.16 ±0.70	87.97 ±1.52	88.36 ±0.33	90.59 ±0.37	76.13 ±0.42
Self-play	SESA	92.66 ±0.43	94.06 ±0.76	81.41 ±0.86	85.55 ±0.90	85.47 ±0.70	92.81 ±0.35	37.81 ±1.05	77.66 ±0.70	87.03 ±0.51	89.81 ±0.59	73.44 ±0.53
	MetaSkill	89.53 ±1.05	92.27 ±1.20	77.97 ±0.86	81.64 ±1.08	78.59 ±0.70	91.56 ±0.65	34.53 ±0.86	72.19 ±1.18	83.75 ±0.21	86.96 ±0.81	69.22 ±0.55
Ours	**R^{2}Orch**	**95.31** ±0.78	**96.17** ±0.89	**85.31** ±0.35	**88.20** ±0.65	**86.56** ±0.86	**94.69** ±0.86	**42.81** ±0.35	**91.09** ±0.70	**90.31** ±0.51	**92.19** ±0.55	**78.79** ±0.17

Table 1: Main results (five-run mean ± std, %), with benchmarks grouped by task domain. All methods share the frozen Workers, the training tasks, and the compute budget, each in its best configuration, and R^{2}Orch is tested with zero skills. Each test set has 128 items (AIME 2026: 30); bold marks the best score in each benchmark column.

5 Experiments

We evaluate R^{2}Orch through the following research questions (RQs). **RQ1:** How does R^{2}Orch compare with the baselines in distribution and on held-out benchmarks? **RQ2:** Does the trained Director transfer to other Worker backbones without retraining? **RQ3:** Do both loops of Recursive Self-Play Orchestration contribute, and does the Director keep its accuracy at zero skills (Section 4.1)? **RQ4:** Does Recursive Learnability choose the tasks that keep yielding a learning signal as skills are withdrawn (Section 4.2)? **RQ5:** Does Skill Withdrawal remove each skill only once the Director no longer needs it (Section 4.3)?

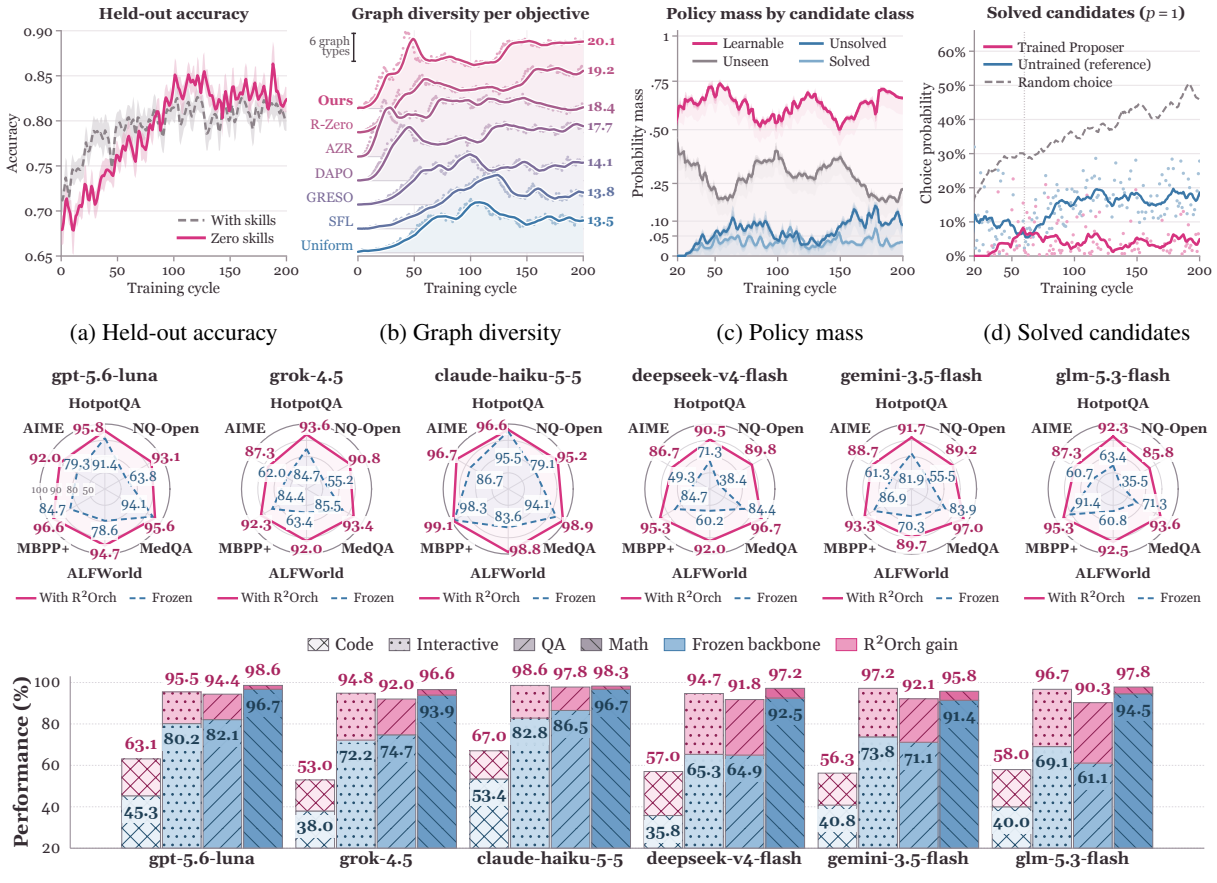
Setup. Six in-distribution (IID) benchmarks supply the training tasks and IID tests, and six out-of-distribution (OOD) benchmarks are held out. The baselines span direct prompting, backbone training, workflow search, RL-trained orchestration, skill internalization, and self-play with skill evolution, under the shared protocol of Table 1 (Appendix D).

Main results (RQ1). R^{2}Orch is best on all twelve benchmarks against six families of baselines, and its lead holds on the held-out half at zero skills (Table 1). Training the backbone alone changes little, while orchestration lifts the frozen Workers far higher, so the gains come from how the Workers are organized. The other orchestrators learn from signals that stop tracking the Director: fixed task pools (workflow search, RL-trained orchestration), success rates or budgets (skill internalization), and frontiers measured with skills in context (self-play). R^{2}Orch replaces these signals with Recursive Learnability and Skill Withdrawal, so its lead is largest on AIME 2026, WebShop, and ALFWorld, where the ablations cost most (Table 2).

Backbone transfer (RQ2). The Director trained with Qwen3.5-9B Workers improves all six other backbones on every benchmark without retraining (Figure 4e), since the roles and hand-offs it builds fit any Worker. Weaker backbones gain the most, narrowing the IID range across backbones from 25.7 to 6.4 points, because the graph supplies the planning and checking they lack. On OOD benchmarks, the mean gain is largest on interactive tasks,

Variant	IID						OOD					
	HotpotQA Ans EM	NQ-Open Ans EM	MedQA Acc.	AIME Acc.	MBPP+ Pass@1	ALFWorld SR	TriviaQA Ans EM	MuSiQue Ans EM	GPQA Acc.	MATH Acc.	SWE Resolved	WebShop SR
Qwen3.5-9B (direct prompting)	57.81	23.59	70.94	48.67	78.59	47.19	44.53	39.22	61.41	89.38	15.78	32.81
Recursive Self-Play Orchestration (Sec. 4.1)												
– Recursive Self-Play Orchestration (GRPO only)	86.09	82.97	88.44	59.33	86.88	73.44	91.41	78.91	80.78	92.03	35.00	71.72
– Skills (no scaffold)	86.88	83.59	89.22	64.00	88.13	77.81	92.34	80.63	82.66	92.34	37.66	76.09
– Self-play (uniform picks)	90.16	83.59	92.34	66.67	90.47	86.88	93.44	81.72	84.53	93.91	38.91	87.03
Recursive Learnability (Sec. 4.2)												
– Recursive Learnability (failure reward)	87.97	82.66	90.47	64.00	90.00	84.06	92.66	81.09	83.44	92.81	38.13	83.13
Learnability on the current scaffold	90.47	84.38	92.03	70.67	91.25	86.88	94.69	83.13	85.47	93.59	40.00	85.94
Dynamic sampling (DAPO)	88.13	83.59	92.50	68.67	91.09	87.50	94.69	82.97	84.84	93.91	39.53	87.81
Skill Withdrawal (Sec. 4.3)												
– Skill Withdrawal (scaffold fixed)	88.13	82.97	90.47	66.67	88.75	77.81	92.34	81.25	83.28	92.81	38.13	76.09
Success-rate threshold (PATS)	90.00	84.53	91.88	74.67	90.31	87.81	93.59	82.97	84.69	94.06	39.53	87.66
Preset budget (SKILL0)	89.69	84.22	91.25	70.67	90.63	86.09	93.75	82.66	84.38	93.75	39.38	85.00
Validation gap (SLIM)	90.63	85.00	93.28	72.67	91.56	88.28	94.84	83.28	85.31	93.59	40.47	89.22
R^{2}Orch (Full)	**91.09**	**86.09**	**94.06**	**81.33**	**93.13**	**90.63**	**95.31**	**85.31**	**86.56**	**94.69**	**42.81**	**91.09**

Table 2: Ablation of R^{2}Orch, tested with zero skills (five-run mean, %). The first row of each group removes the whole component and the other rows change one part (Appendix D); free-form QA benchmarks report Ans EM, and the Qwen3.5-9B and R^{2}Orch rows repeat their scores in Table 1.



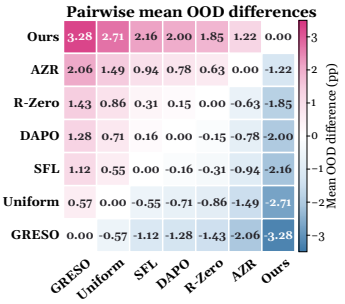
(a) Held-out accuracy (b) Graph diversity (c) Policy mass (d) Solved candidates

(e) Backbone transfer: IID (radars) and OOD (bars)

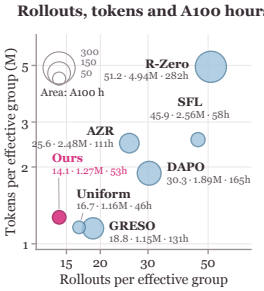
Figure 4: Training dynamics and backbone transfer. (a) Held-out accuracy with skills and at zero skills. (b) Graph types over the six IID datasets per Proposer objective (final counts at the curve ends). (c) The Proposer's probability mass on unseen, unsolved ($p = 0$), learnable ($0 < p < 1$), and solved ($p = 1$) candidates by previous success rate p (power-scaled axis). (d) Probability of choosing a solved candidate as a moving average over 10 cycles (dots: single cycles) for the trained Proposer, the untrained model with the same prompt, and random choice. (e) Backbone transfer: the Director trained with Qwen3.5-9B Workers is used unchanged at zero skills with six other backbones. Radars: IID scores with R^{2}Orch (solid) and without it (dashed), the radius linear over 20–80 (inner half) and 80–100 (outer half). Bars: OOD scores per task domain, the frozen backbone in blue and R^{2}Orch's gain in pink (Code: SWE-Bench; Interactive: WebShop; QA: mean of TriviaQA, MuSiQue, and GPQA; Math: MATH-Hard).

Proposer objective	HotpotQA	NQ-Open	MedQA	AIME	MBPP+	ALFWorld
	Ans EM	Ans EM	Acc.	Acc.	Pass@1	SR
Uniform sampling	90.16	83.59	92.34	66.67	90.47	86.88
Dynamic sampling (DAPO)	88.13	83.59	92.50	68.67	91.09	87.50
Selective rollout (GRESO)	88.44	83.28	91.56	68.00	90.31	85.78
Learnability, current scaffold (SFL)	90.47	84.38	92.03	70.67	91.25	86.88
Solve-rate reward (Absolute Zero)	90.78	84.53	92.34	77.33	92.19	88.13
Uncertainty reward (R-Zero)	90.31	84.38	92.50	70.00	91.72	87.34
Recursive Learnability (ours)	**91.09**	**86.09**	**94.06**	**81.33**	**93.13**	**90.63**

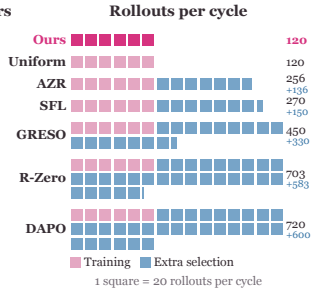
(a) IID scores per Proposer objective



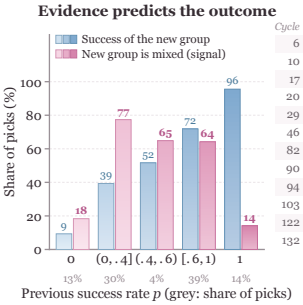
(b) Pairwise OOD differences



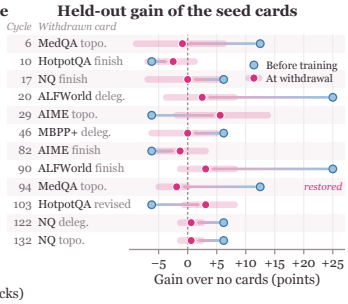
(c) Per effective group



(d) Per cycle



(e) Evidence and outcome



(f) Skill withdrawals

Figure 5: Proposer objectives, selection cost, and skill withdrawals (RQ4, RQ5); only the Proposer's objective changes in (a)–(d) (AZR: Absolute Zero). (a) IID scores. (b) Row minus column of the mean OOD score. (c) Rollouts and tokens per effective group; bubble area: A100 hours. (d) Training (pink) and selection (blue) rollouts per cycle. (e) Seen-task picks over cycles 50–199 by previous success rate p: realized success and share of mixed new groups. (f) The seed cards' held-out gain over no cards, before training and around each withdrawal (95% CI).

whose many steps compound each hand-off, and smallest on math, which is already near its ceiling.

Recursive Self-Play Orchestration (RQ3). Both loops are needed (Table 2): skills give the Director the early successes that make hard tasks learnable, and the Proposer keeps groups away from uniform outcomes. Since most of each group hides the candidate skill (Eq. (4)), updates teach the Director to succeed without it, and zero-skill accuracy overtakes the accuracy with skills midway (Figure 4a). A skill has thus stopped helping before it leaves, keeping the outer costs of Eq. (6) small as the inner gains of self-play accumulate (Proposition 1).

Recursive Learnability (RQ4). Recursive Learnability picks the tasks that Proposition 2 expects to gain most, those whose rollouts still disagree with the candidate skill withheld. New groups on tasks of intermediate success rate reach mixed outcomes, and thus a nonzero GRPO signal, about four times as often as at $p = 0$ or $p = 1$ (Figure 5e), and the trained Proposer learns to skip solved candidates (Figure 4c,d). Learnability measured with the skill in context picks tasks that collapse once the skill leaves, so it trails on every IID benchmark and the mean OOD score, as does every other objective (Table 2, Figure 5a,b). Since the hidden part already measures this score, selection costs no extra rollouts and the fewest per effective group (Figure 5c,d), and these groups lead the Director to build the most distinct graph types (Figure 4b).

Skill Withdrawal (RQ5). Skill Withdrawal removes a skill once it no longer helps the current Director, as it reads the skill's own effect, not the success rate. The success rate mixes that effect with the curriculum that Recursive Learnability shapes (Eq. (12)), so a success-rate threshold misreads the skill and trails the paired contrast, as do a preset budget and a validation gap (Proposition 3, Table 2). All 12 withdrawals pass Eq. (14) on the card's own contrasts, with posterior $\mathbb{E}[\Delta^+]$ at most 0.019, and around each the seed cards no longer raise held-out accuracy (Figure 5f). A fixed scaffold costs most at zero skills, as the Director keeps leaning on skills that never leave its context throughout training.

6 Conclusion

R^{2}Orch nests two loops: Recursive Learnability picks tasks learnable without the candidate skill, and Skill Withdrawal retires skills that stop helping. It leads on all twelve benchmarks at zero skills.

Limitations

R^{2}Orch starts from initial skills that we customize by hand for each dataset, and the withdrawals, revisions, and distillations that follow build on them, so deriving the initial skills automatically for a new domain remains an open problem.

Ethical considerations

R^{2}Orch is trained and evaluated on public benchmarks that we use for research evaluation as their creators intend, and the study involves no human participants, annotators, or personal data. ALFWorld and WebShop are simulated environments, and the code benchmarks run in their standard evaluation harnesses. Agents that orchestrate tools can be misused like any agent system, and a self-improving orchestrator shares this risk, so we release the code for inspection and recommend running it in a sandboxed environment.

Acknowledgements

We used AI assistants to polish the language and to help with LaTeX typesetting; all other work, including the experiments and the first drafts, is the authors' own, and the authors take full responsibility for the content of the paper.

References

Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, and Sara Hooker. 2024. Back to basics: Revisiting reinforce-style optimization for learning from human feedback in llms. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12248–12267.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. 2021. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

James O. Berger. 1985. *Statistical Decision Theory and Bayesian Analysis*. Springer New York.

Vivek S Borkar. 1997. Stochastic approximation with two time scales. *Systems & Control Letters*, 29(5):291–294.

Michael Dennis, Natasha Jaques, Eugene Vinitsky, Alexandre Bayen, Stuart Russell, Andrew Critch, and Sergey Levine. 2020. Emergent complexity and zero-shot transfer via unsupervised environment design. *Advances in neural information processing systems*, 33:13049–13061.

Zenghuang Fu, Zhaoyang Li, Qiuyuan Ai, Haoyu Wu, Minghui Wu, Chenxu Zhao, Ante Wang, Guannan He, and Changwei Wang. 2026. Self-play meets skill evolution: Self-evolving search agents that pose, solve, and remember. *arXiv preprint arXiv:2607.29468*.

Andrew Gelman, John B Carlin, Hal S Stern, and Donald B Rubin. 1995. *Bayesian data analysis*. chapman and hall/cRc.

Xixiang He, Qiyao Sun, Ao Cheng, Xingming Li, Xuanyu Ji, Hailun Lu, Runke Huang, and Qingyong Hu. 2026. Advantage collapse in group relative policy optimization: Diagnosis and mitigation. *arXiv preprint arXiv:2605.21125*.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*.

Chengsong Huang, Wenhao Yu, Xiaoyang Wang, Hongming Zhang, Zongxia Li, Ruosen Li, Jiaxin Huang, Haitao Mi, and Dong Yu. 2026. R-zero: Self-evolving reasoning llm from zero data. In *International Conference on Learning Representations*, volume 2026, pages 130770–130790.

Guido W Imbens and Donald B Rubin. 2015. Causal inference in statistics, social, and biomedical sciences. *New York*, 517.

Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. Swe-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations*, volume 2024, pages 54107–54157.

Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2021. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421.

Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611.

Tomasz Korbak, Ethan Perez, and Christopher Buckley. 2022. Rl with kl penalties is better viewed as bayesian inference. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 1083–1091.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466.

Kenton Lee, Ming-Wei Chang, and Kristina Toutanova. 2019. Latent retrieval for weakly supervised open domain question answering. In *Proceedings of the 57th annual meeting of the association for computational linguistics*, pages 6086–6096.

Zhuofeng Li, Haoxiang Zhang, Seungju Han, Sheng Liu, Jianwen Xie, Yu Zhang, Yejin Choi, James Y Zou, and Pan Lu. 2026. In-the-flow agentic system optimization for effective planning and tool use. In *International Conference on Learning Representations*, volume 2026, pages 50524–50570.

Hongxiang Lin, Zhirui Kuai, Erpeng Xue, and Lei Wang. 2026. Skillc: Learning autonomous skill internalization in llm agents via contrastive credit assignment. *arXiv preprint arXiv:2605.27899*.

Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. 2023. Is your code generated by chatgpt really correct? rigorous evaluation of large language models for code generation. *Advances in neural information processing systems*, 36:21558–21572.

Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. 2025. Understanding r1-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*.

Shuqi Lu, Chaofan Li, Kun Luo, Zhang Zhang, Hui Wang, Hongwang Xiao, Lei Xiong, Jiahao Wang, Sen Wang, Xiyan Jiang, et al. 2026a. Arex: Towards a recursively self-improving agent for deep research. *arXiv preprint arXiv:2607.21461*.

Zhengxi Lu, Zhiyuan Yao, Jinyang Wu, Chengcheng Han, Qi Gu, Xunliang Cai, Weiming Lu, Jun Xiao, Yueting Zhuang, and Yongliang Shen. 2026b. Skill0: In-context agentic reinforcement learning for skill internalization. *arXiv preprint arXiv:2604.02268*.

Jan Peters, Katharina Mulling, and Yasemin Altun. 2010. Relative entropy policy search. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 24, pages 1607–1612.

Qwen Team. 2026. Qwen3.5: Towards native multimodal agents.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. 2023. Gpqa: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*.

Donald B Rubin. 1974. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology*, 66(5):688.

Donald B Rubin. 1981. The bayesian bootstrap. *The annals of statistics*, pages 130–134.

Alex Rutherford, Michael Beukman, Timon Willi, Bruno Lacerda, Nick Hawes, and Jakob Foerster. 2024. No regrets: Investigating and improving regret approximations for curriculum discovery. *Advances in Neural Information Processing Systems*, 37:16071–16101.

Jürgen Schmidhuber. 2007. Gödel machines: Fully self-referential optimal universal self-improvers. In *Artificial general intelligence*, pages 199–226. Springer.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Zayx Shawn. 2026. Pace: Anytime-valid acceptance tests for self-evolving agents. *arXiv preprint arXiv:2606.08106*.

Junhao Shen, Teng Zhang, Xiaoyan Zhao, and Hong Cheng. 2026. Dynamic skill lifecycle management for agentic reinforcement learning. *arXiv preprint arXiv:2605.10923*.

Yipeng Shi, Zhipeng Ma, Yue Wang, Qitai Tan, Yang Li, Peng Chen, and Zhengzhou Zhu. 2026. Pats: Policy-aware training scaffolding for agentic reinforcement learning. *arXiv preprint arXiv:2607.21419*.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2020. Alfworld: Aligning text and embodied environments for interactive learning. *arXiv preprint arXiv:2010.03768*.

Dhruv Srikanth, Bingchen Zhao, Dixing Xu, Yuxiang Wu, and Zhengyao Jiang. 2026. Recursive self-improvement of ai research agents. *arXiv preprint arXiv:2609.26457*.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. ♪ musique: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Songjun Tu, Chengdong Xu, Qichao Zhang, Yaocheng Zhang, Xiangyuan Lan, Linjing Li, Dong Li, and Dongbin Zhao. 2026. Dynamic dual-granularity skill bank for agentic rl. *arXiv preprint arXiv:2603.28716*.

Zefeng Wang, Minxi Yan, Jinhe Bi, Sikuan Yan, Volker Tresp, and Yunpu Ma. 2026. Metaskill-evolve: Recursive self-improvement of llm agents via two-timescale meta-skill evolution. *arXiv preprint arXiv:2607.05297*.

R Thompson William et al. 1933. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika*, 25(3/4):285–294.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pages 2369–2380.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. Webshop: Towards scalable real-world web interaction with grounded language agents. *Advances in Neural Information Processing Systems*, 35:20744–20757.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. 2026. Dapo: An open-source llm reinforcement learning system at scale. *Advances in Neural Information Processing Systems*, 38:113222–113244.

Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. 2026a. Darwin gödel machine: Open-ended evolution of self-improving agents. In *International Conference on Learning Representations*, volume 2026, pages 104223–104294.

Jenny Zhang, Bingchen Zhao, Wannan Yang, Jakob Foerster, Jeff Clune, Minqi Jiang, Sam Devlin, and Tatiana Shavrina. 2026b. Hyperagents. *arXiv preprint arXiv:2603.19461*.

Jiayi Zhang, Jinyu Xiang, Zhaoyang Yu, Fengwei Teng, Xionghui Chen, Jiaqi Chen, Mingchen Zhuge, Xin Cheng, Sirui Hong, Jinlin Wang, et al. 2025. Aflow: Automating agentic workflow generation. In *International Conference on Learning Representations*, volume 2025, pages 34040–34077.

Mingda Zhang, Wenjin Liu, Tiesunlong Shen, Qika Lin, Rui Mao, Erik Cambria, Xiaoying Tang, and Haoran Luo. 2026c. Flowsteer: Towards agents designing agentic workflows via reinforced progressive canvas editing. *arXiv preprint arXiv:2602.01664*.

Mingda Zhang, Tiesunlong Shen, Haoran Luo, Wenjin Liu, Zikai Xiao, Erik Cambria, and Xiaoying Tang. 2026d. Skillflow: Flow-driven recursive skill evolution for agentic orchestration. *arXiv preprint arXiv:2605.14089*.

Xing Zhang, Yanwei Cui, Guanghui Wang, Ziyuan Li, Wei Qiu, Bing Zhu, and Peiyang He. Ratchet: How reliable must an llm judge be to retire a skill?, 2026d. *URL https://arxiv. org/abs/2605.22148*.

Xing Zhang, Yanwei Cui, Guanghui Wang, Ziyuan Li, Wei Qiu, Bing Zhu, and Peiyang He. 2026e. Library drift: Diagnosing and fixing a silent failure mode in self-evolving llm skill libraries. *arXiv preprint arXiv:2605.19576*.

Andrew Zhao, Yiran Wu, Tong Wu, Quentin Xu, Yang Yue, Matthieu Lin, Shenzhi Wang, Qingyun Wu, Zilong Zheng, and Gao Huang. 2026. Absolute zero: Reinforced self-play reasoning with zero data. *Advances in Neural Information Processing Systems*, 38:105816–105879.

Haizhong Zheng, Yang Zhou, Brian Bartoldson, Bhavya Kailkhura, Fan Lai, Jiawei Zhao, and Beidi Chen. 2026. Act only when it pays: Efficient reinforcement learning for llm reasoning via selective rollouts. *Advances in Neural Information Processing Systems*, 38:124321–124346.

Sibo Zhu, Shicheng Fan, Xinyue Wang, Wenyi Wu, Kun Zhou, and Biwei Huang. 2026. Rsiagent: Autonomous exploration for recursive self-improvement in new environments. *arXiv preprint arXiv:2609.15364*.

A Theory and Proofs

This appendix analyzes the two channels of R^{2}Orch: outcome diversity for the curriculum and randomized evidence for skill withdrawal. We derive the policy-space progress calculation, the rollout estimators, and the posterior decisions, then connect them through round accounting. Proposition 1 rests on Lemmas A.4–A.6, Proposition 2 on Lemmas A.1–A.3 and A.9, and Proposition 3 on Lemmas A.7 and A.8; the final subsection collects the three proofs, and Table 3 the notation.

Symbol	Meaning
$\pi_\theta, \rho_\phi, \rho_0$	Director, Proposer, Proposer reference
ρ, C_t	evaluation distribution, candidate list
$\mathcal{S}_t, K_x, k_x$	scaffold, skills retrieved for x, skill under test
$G = n_h + n_v$	rollouts per group (hidden, visible)
$\mathcal{H}_x, \mathcal{V}_x$	parts reading $K_x \setminus k_x$ and K_x
y_i, u_i	correctness; correctness less Worker cost
p^+, p^-	success probability under K_x, $K_x \setminus k_x$
$\ell(p), \ell_{\text{rec}}$	learnability $4p(1-p)$; its value at p^-
$R_P, \tilde{p}_x$	Proposer reward; shrunk success rate
$z_x, \Delta_k(x)$	paired contrast; skill effect on task x
Π_k, Π_k^0	posterior of Δ_k; with one prior record
$\mathcal{W}_t, \mathcal{N}_t$	skills withdrawn, admitted in round t
$J_{\mathcal{S}}, \Gamma_t, C_t$	success rate under $\mathcal{S}$; inner gain; outer cost
β, τ	Director step size; Proposer KL weight

Table 3: Notation of the main text.

A.1 Conditional setup and policy likelihoods

Fix a task x, a scaffold, a retrieved context K, and a candidate skill $k \in K$. Write $K^- = K \setminus \{k\}$, $p^+ = p_K(x;\theta)$, and $p^- = p_{K^-}(x;\theta)$. The success probability averages both Director sampling and Worker execution. The two parts have fixed positive sizes n_h, n_v, with $G = n_h + n_v$. During one group, the policy, candidate, contexts, and pre-group statistics are held fixed. Conditional on the assigned contexts, different rollouts use independent randomness. Task rewards average candidate-skill selection whenever several skills can be tested for the same task. Gradients hold the task and context sampling laws fixed and assume the usual measurability and integrability conditions.

For an execution that returns intermediate Worker feedback, let ω denote its complete record and h_t the history before Director action a_t. Frozen Worker kernels are independent of θ, so the likelihood score is

$$S_\theta(\omega) = \nabla_\theta \log P_\theta(\omega) = \sum_t \nabla_\theta \log \pi_\theta(a_t \mid h_t). \tag{15}$$

Thus $\mathbb{E}[S_\theta] = 0$ and $\nabla_\theta \mathbb{E}[u] = \mathbb{E}[u S_\theta]$ for a reward with no direct parameter dependence. Multiple editing histories may produce the same terminal graph; their individual action likelihoods are not the marginal likelihood of that graph. The group-gradient results use the record score in Eq. (15). A terminal-graph score gives the same value gradient when the conditional reward mean given that graph is independent of θ. Otherwise the complete record is required, for example when editing histories with different costs reach the same graph.

The policy-space tilt below considers a choice g followed by a fixed execution kernel, so its stochastic extension leaves Worker randomness outside the Director's control. We use a fixed evaluation distribution ρ for round accounting. The training curriculum ρ_ϕ and the distribution of tasks recorded for a particular skill need not equal ρ.

In Eq. (3), the Director KL is understood as averaged over the fixed task and context distribution. Its moving reference π_{θ_t} defines an ideal proximal response; the base-model anchor in the training loss is a separate regularizer. The Proposer instead uses the fixed reference ρ_0 in both its KL term and the trust region on its pair tilt (Appendix B), so its response is proximal to ρ_0.

A.2 KL tilting and controllable progress

Lemma A.1 (KL-tilt progress)**.** *Let π be a reference distribution and $F(g)$ a bounded, parameter-independent score. For $\beta > 0$, the unique maximizer, on the support of π, of $\mathbb{E}_{\pi'}[F] - \beta^{-1}\mathrm{KL}(\pi'\|\pi)$ is*

$$\pi_\beta(g) = \frac{\pi(g)e^{\beta F(g)}}{Z(\beta)}, \qquad Z(\beta) = \mathbb{E}_\pi[e^{\beta F}]. \tag{16}$$

Writing $\psi = \log Z$, $\mu = \mathbb{E}_\pi F$, and $V = \mathrm{Var}_\pi(F)$ gives

$$\begin{aligned} \psi(\beta) &= \beta\mu + \tfrac{1}{2}\beta^2 V + O(\beta^3), \\ \mathbb{E}_{\pi_\beta} F - \mu &= \beta V + O(\beta^2), \\ \mathrm{KL}(\pi_\beta\|\pi) &= \tfrac{1}{2}\beta^2 V + O(\beta^3). \end{aligned} \tag{17}$$

For fixed $V > 0$, a small KL budget δ yields optimal score gain $\sqrt{2\delta V} + O(\delta)$.

Proof. Substitution of the tilted density gives, for every admissible π',

$$\mathbb{E}_{\pi'} F - \beta^{-1}\mathrm{KL}(\pi'\|\pi) = \beta^{-1}\{\log Z - \mathrm{KL}(\pi'\|\pi_\beta)\}. \tag{18}$$

Nonnegativity of KL proves optimality and uniqueness. Boundedness permits differentiation of Z: $\psi'(\beta) = \mathbb{E}_{\pi_\beta} F$ and $\psi''(\beta) = \mathrm{Var}_{\pi_\beta}(F)$. Expanding at zero proves the first two identities. Since the log density ratio is $\beta F - \psi(\beta)$, its expectation is $\beta\psi'(\beta) - \psi(\beta)$, proving the KL expansion. For $V > 0$, inversion gives $\beta = \sqrt{2\delta/V} + O(\delta)$. The variational inequality also shows that this tilt maximizes expected score among distributions with its KL cost. Substitution proves the budget statement. If $V = 0$, the score is constant on the available support and no such tilt improves it. □

For a binary outcome determined by g, take $F = r$ and $p = \mathbb{E}_\pi r$. Then

$$Z(\beta) = 1 - p + pe^\beta, \qquad p_\beta = \frac{pe^\beta}{Z(\beta)}, \\ p_\beta - p = \frac{p(1-p)(e^\beta - 1)}{1 + p(e^\beta - 1)}. \tag{19}$$

Consequently $\psi''(0) = p(1-p)$ is also the Fisher information of this one-dimensional exponential family. At a common small tilt, gains rank by $p(1-p)$ to first order; at a common small KL budget, they scale with its square root. At a finite tilt the ranking can change: at $\beta = \log 2$, $p = 0.3$ yields gain $0.21/1.3$, whereas $p = 0.6$ yields the smaller gain $0.24/1.6$ despite its larger variance. Gain per unit KL cost is $2/\beta + O(1)$.

Stochastic Workers. Let $Q(g) = \mathbb{E}[Y \mid g]$ for a fixed Worker execution kernel and $Y \in \{0, 1\}$. Optimizing expected correctness over graph distributions applies Lemma A.1 with $F = Q$, not with a sampled Y. The controllable graph-level variance is

$$V_{\mathrm{ctrl}} = \mathrm{Var}_\pi(Q), \\ p(1-p) = V_{\mathrm{ctrl}} + \mathbb{E}_\pi[Q(1-Q)]. \tag{20}$$

The second identity follows by conditioning $Y^2 = Y$ on g, or by total variance. A Director-only update reweights graphs and keeps the Worker kernel, whereas a joint tilt of graph and execution outcomes by $e^{\beta Y}$ would change that kernel. In particular, if $Q(g) = 1/2$ for every graph, correctness variance is $1/4$ but no graph redistribution improves success. Recursive Learnability measures outcome diversity under stochastic execution, which equals its controllable part for graph-determined outcomes, the setting of Proposition 2.

A local statement also covers feedback-dependent records and restricted policy classes. Let $b = \mathbb{E}[(Y - p)S_\theta]$ and $I = \mathbb{E}[S_\theta S_\theta^\top]$. Under a differentiable local KL expansion, maximizing $b^\top d$ subject to $d^\top I d/2 \le \delta$ gives

$$\sqrt{2\delta\, b^\top I^\dagger b}, \qquad b^\top I^\dagger b \le p(1-p). \tag{21}$$

Indeed, a null direction of I has zero score almost surely and is orthogonal to b. Cauchy–Schwarz in the range of I gives the optimum. Moreover $S_\theta^\top I^\dagger b$ is the least-squares projection of $Y - p$ onto the score span, so its variance cannot exceed $\mathrm{Var}(Y)$. This identifies the accessible part of outcome diversity for any differentiable policy class.

For the cost-adjusted reward $u = Y - \lambda c$, define $U(g) = Q(g) - \lambda\mathbb{E}[c \mid g]$. Tilting by U changes correctness at the initial rate $\mathrm{Cov}_\pi(Q, U) = \mathrm{Var}_\pi(Q) - \lambda\,\mathrm{Cov}_\pi(Q, \bar{c})$ with $\bar{c}(g) = \mathbb{E}[c \mid g]$, which differs from $\mathrm{Var}_\pi(Q)$ by at most $\lambda\,(\mathrm{Var}_\pi(Q)\,\mathrm{Var}_\pi(\bar{c}))^{1/2}$, small for the weight λ of Table 4.

Lemma A.2 (Equal tilt)**.** *For finitely many tasks with fixed weights $a_j > 0$ and variances $V_j > 0$, the quadratic-budget problem*

$$\max_{\beta_j \ge 0} \sum_j a_j \beta_j V_j \quad \textit{subject to} \quad \tfrac{1}{2}\sum_j a_j \beta_j^2 V_j \le \delta \tag{22}$$

has the common solution $\beta_j = \sqrt{2\delta/\sum_j a_j V_j}$.

Proof. Cauchy–Schwarz bounds the squared objective by $(\sum_j a_j\beta_j^2 V_j)(\sum_j a_j V_j)$, with equality exactly when all positive-variance tasks have the same tilt. Saturating the budget gives the stated value. Zero-variance tasks add no gain or cost. □

This allocation result for independently adjustable policy distributions motivates one advantage scale across tasks: the batch whitening of Appendix B rescales every task by the same factor, whereas dividing by each task's standard deviation would give larger multipliers to low-variance tasks.

A.3 Advantage energy, collapse, and shrinkage

Lemma A.3 (Advantage energy)**.** *Let $n \ge 2$ independent outcomes have distribution* Bernoulli(p) *and let $A_i = Y_i - \bar{Y}$. Then*

$$\mathbb{E}\sum_i A_i^2 = (n-1)p(1-p), \\ \Pr\left(\sum_i A_i^2 = 0\right) = p^n + (1-p)^n. \tag{23}$$

Thus $\ell(p) = 4p(1-p)$ is $4/(n-1)$ times the expected advantage energy, and a realized group collapses exactly when its outcomes agree.

Proof. For $B = \sum_i Y_i$, binary outcomes give $\sum_i A_i^2 = B - B^2/n$. Since $\mathbb{E}B = np$ and $\mathbb{E}B^2 = np(1-p) + n^2p^2$, taking expectations proves the first identity. The sum vanishes exactly at $B \in \{0, n\}$, giving the second identity. □

For the actual hidden part, $n = n_h$, and the hypothetical homogeneous group in Eq. (8) has $n = G$. Population learnability vanishes only at $p \in \{0, 1\}$, while a finite group can collapse at any p: even at $p = 1/2$, with probability 2^{1-n}. For $n > 1$, differentiating the collapse probability gives $n[p^{n-1} - (1-p)^{n-1}]$, which is negative below one half and positive above it. Thus the same population frontier that maximizes learnability minimizes homogeneous-group collapse probability. The hidden part and the full group consequently have different finite-sample collapse probabilities even when their success probabilities coincide.

The mixed group has a different energy. Let $V_- = p^-(1-p^-)$ and $V_+ = p^+(1-p^+)$. Conditional independence gives

$$\mathbb{E}\sum_i (Y_i - \bar{Y})^2 = \left(1 - \tfrac{1}{G}\right)(n_h V_- + n_v V_+) + \tfrac{n_h n_v}{G}(p^+ - p^-)^2. \tag{24}$$

To verify this, expand $\sum_i Y_i^2 - G\bar{Y}^2$ and separate within-context variances from the squared differences between context means. Its collapse probability is $(p^-)^{n_h}(p^+)^{n_v} + (1-p^-)^{n_h}(1-p^+)^{n_v}$. The extra energy term measures dependence on the skill rather than post-withdrawal learning potential, which is why Eq. (9) reads the hidden part. If advantages use u, the same expansion concerns reward variance, and correctness agreement need not make them zero when costs differ.

Meaning of the Beta reward. Condition on a pre-group running mean $\bar{p} \in (0, 1)$ and strength $m > 0$. A proper working prior is $p \sim \mathrm{Beta}(a, b)$ with $a = m\bar{p}$, $b = m(1-\bar{p})$. For a fixed context with s successes in n independent trials, set $a' = a + s$, $b' = b + n - s$, and $N = n + m$. Multiplication of prior and Bernoulli likelihood gives the posterior $\mathrm{Beta}(a', b')$, with

$$\tilde{p} = a'/N, \qquad \mathrm{Var}(p \mid y) = \frac{\tilde{p}(1-\tilde{p})}{N+1},$$
$$\mathbb{E}[\ell(p) \mid y] = \frac{N}{N+1}\ell(\tilde{p}). \tag{25}$$

The last identity follows from $\mathbb{E}[p(1-p)] = a'b'/[N(N+1)]$. Hence Eq. (9) uses the posterior-mean *plug-in*, which exceeds the posterior expectation of learnability by $4\mathrm{Var}(p \mid y)$. A running mean estimated from data gives an empirical-Bayes interpretation conditional on that mean.

The plug-in also has a precise frequentist interpretation when the running mean is fixed before sampling. Let $\mu_n = (np + m\bar{p})/(n+m)$. Since $\mathbb{E}\tilde{p} = \mu_n$ and $\mathrm{Var}(\tilde{p}) = np(1-p)/(n+m)^2$, expanding the quadratic learnability function gives

$$\mathbb{E}[\ell(\tilde{p})] = \ell(\mu_n) - \frac{4np(1-p)}{(n+m)^2}. \tag{26}$$

There are therefore two separate effects: shrinkage changes the center, while finite sampling subtracts a variance term. The identity holds before the outcome-dependent branch selection below.

For paired groups, the stated branching can be written as follows: c_x indicates agreement of all G outcomes; on a noncollapsed group, $h_x = 1$ when the hidden part has no success and the visible part does. Use $\mathcal{A}_x = \mathcal{V}_x$ in this transfer branch and $\mathcal{A}_x = \mathcal{H}_x$ otherwise. A no-skill group uses all its rollouts. The visible branch estimates visible-context success p^+ and supplies a discounted transfer incentive. The collapse penalty and the compute factor further shape the operational reward, whose expectation $\bar{R}_P$ enters Eq. (10).

There is nevertheless a direct large-group connection in the ordinary hidden-success branch. Fix $0 < p^- < 1$, let both part sizes increase, and take fixed m or $m/n_h \to 0$. Consistency of the hidden sample mean gives $\tilde{p}_x \to p^-$ in probability. The hidden-zero event has probability $(1-p^-)^{n_h} \to 0$. Full-group collapse is contained in the event that the hidden outcomes all agree, whose probability also tends to zero. Therefore, with the compute factor omitted, the bounded reward in Eq. (9) converges to $\ell(p^-)$ in probability and in expectation.

The exceptional branch still matters. At $p^- = 0$ and $0 < p^+ < 1$, the hidden part keeps failing while the visible part includes a success with high probability. With a consistent visible estimate, the reward approaches $\gamma\ell(p^+)$, so the transfer term

keeps a discounted incentive for assisted practice on tasks whose hidden-context learnability is zero.

A.4 Mixed-context gradients and the last skill

Lemma A.4 (Mixed-group gradient)**.** *Condition on an assignment of exactly n_h hidden and n_v visible contexts. Let S_i be each rollout's likelihood score and u_i its integrable reward. With independent rollouts conditional on this assignment and a common group mean,*

$$\mathbb{E}\left[\frac{\sum_i (u_i - \bar{u})S_i}{G-1}\right] = \frac{n_h}{G}\nabla v_{K^-} + \frac{n_v}{G}\nabla v_K. \quad (27)$$

If each part is centered separately, the expectation of the unnormalized sum becomes

$$(n_h - 1)\nabla v_{K^-} + (n_v - 1)\nabla v_K. \quad (28)$$

Proof. Write $\mathbf{K}$ for the complete context assignment. For $j \neq i$, conditional independence and the zero-mean score identity imply $\mathbb{E}[u_j S_i \mid \mathbf{K}] = \mathbb{E}[u_j \mid \mathbf{K}]\mathbb{E}[S_i \mid \mathbf{K}] = 0$. For $j = i$, $\mathbb{E}[u_i S_i \mid \mathbf{K}] = \nabla v_{K_i}$. Therefore

$$\mathbb{E}[(u_i - \bar{u})S_i \mid \mathbf{K}] = \left(1 - \tfrac{1}{G}\right)\nabla v_{K_i}. \quad (29)$$

Summing and dividing by $G - 1$ proves Eq. (27). It also follows from the exact leave-one-out identity $u_i - \bar{u} = (1 - 1/G)(u_i - \bar{u}_{-i})$. Repeating the calculation within a part of size n_c gives a total $(n_c - 1)\nabla v_c$, proving Eq. (28). Averaging over assignments preserves both conclusions. □

If the last-skill case $K = \{k\}$ uses separate centering, normalizing Eq. (28) by $G - 2$ gives the mixture

$$q'_x = \frac{n_h - 1}{G-2}\delta_\varnothing + \frac{n_v - 1}{G-2}\delta_{\{k\}}. \quad (30)$$

For $G > 2$ and $n_h > n_v \geq 1$, its hidden weight exceeds n_h/G by $(n_h - n_v)/[G(G-2)]$. Keeping the divisor $G - 1$ instead gives $(G-2)/(G-1)$ times this mixture gradient: the $3/2$ split yields weights $2:1$ and relative scale $3/4$. A singleton part contributes no centered gradient. Once no skill is retrieved, all G rollouts share the empty context and ordinary whole-group centering applies.

These identities describe the on-policy reward gradient; PPO clipping, base-model anchoring, batch whitening, token averaging, and advantage-based filtering turn it into the training surrogate of Appendix B on the same context mixture.

A.5 Round accounting and interacting withdrawals

Lemma A.5 (Round decomposition)**.** *Fix ρ and define $J_S(\theta) = \mathbb{E}_{x\sim\rho}[p_{K_x(S)}(x;\theta)]$. For any finite sequence of policies and scaffolds, define Γ_t and C_t as in Eq. (6). Then*

$$J_{S_T}(\theta_T) = J_{S_0}(\theta_0) + \sum_{t<T}(\Gamma_t - C_t). \quad (31)$$

If an outer change removes distinct skills $k_1, \ldots, k_m$ without changing or refilling the remaining retrieved context, put $K_x^j = K_x \setminus \{k_1, \ldots, k_j\}$. At θ_{t+1} its cost is

$$C_t = \sum_{j=1}^{m} \mathbb{E}_\rho\left[p_{K_x^{j-1}}(x) - p_{K_x^j}(x)\right]. \quad (32)$$

A term is zero on tasks not retrieving k_j.

Proof. Add and subtract $J_{S_t}(\theta_{t+1})$ in each round's total change to obtain $\Gamma_t - C_t$, and telescope over rounds. For pure removals, telescope again over $K_x^0, \ldots, K_x^m$ before taking the task expectation. Both calculations are algebraic, so they hold for any sequence of policies and scaffolds. □

The sequential effects in Eq. (32) can differ from single-removal effects. For two redundant skills a, b with success probabilities $p_{\{a,b\}} = p_{\{a\}} = p_{\{b\}} = 1$ and $p_\varnothing = 0$, each single removal costs zero, but removing both costs one. Bounds therefore add over the successive contexts; in our runs no two withdrawals share a round (Figure 5f), and retesting withdrawn skills (Appendix C) can restore a skill whose removal hurts. Admission, revision, and retrieval replacement enter through Eq. (31).

For one skill, let $a_k = \Pr_\rho(k \in K_x)$ and let $\mu_{\rho,k}$ be its mean current effect conditional on retrieval. Then $C_t = a_k \mu_{\rho,k}$. A ledger instead targets the tasks on which that skill was selected for testing. These distributions can differ even when the task curriculum is fixed, because skill selection depends on the task and ledger. They can also differ from the fixed evaluation distribution in Eq. (31).

The posterior bound transfers to the round cost up to a drift d_k: if a posterior concerns μ_k, the desired current mean satisfies $\mu_{\rho,k} \leq \mu_k + d_k$ under the same model, and withdrawal gives $\mathbb{E}[\mu_k^+] \leq \epsilon$, then

$$\mathbb{E}[C_t] \leq a_k(\epsilon + d_k). \quad (33)$$

Matched policy, context, and task scope suffice for $d_k = 0$. For effects in $[-1, 1]$, a uniform effect drift

b_k and a task-distribution mismatch contribute at most $b_k + 2\mathrm{TV}(\rho(\cdot \mid k \in K_x), \nu_k)$, where ν_k is the distribution of the tasks that test k. This follows by adding and subtracting the old effect under the new distribution and using the total-variation bound.

When several withdrawals are justified on successive contexts, let μ_j denote each retrieval-conditional mean effect and a_j its evaluation retrieval mass. Under a common posterior model with $\mathbb{E}[\mu_j^+] \le \epsilon_j$ for each actual successive effect, linearity gives $\mathbb{E}[C_t] \le \sum_j a_j \epsilon_j$. Independence between the effects is unnecessary; the marginals only need to refer to the successive contexts.

The fixed evaluation distribution also matters. For changing distributions ρ_t, define inner gains and outer costs using ρ_t within round t. Writing J^ν for evaluation under ν, an additional term is

$$D_t = J^{\rho_{t+1}}_{\mathcal{S}_{t+1}}(\theta_{t+1}) - J^{\rho_t}_{\mathcal{S}_{t+1}}(\theta_{t+1}). \tag{34}$$

Adding and subtracting the second expectation at every transition yields a telescoping sum of $\Gamma_t - C_t + D_t$, so changing ρ_t alters measured success without changing the Director. Keeping ρ fixed in the deployed target removes D_t, while the Proposer still chooses training tasks adaptively.

A.6 Tracking under separated scaffold changes

Lemma A.6 (Two-time-scale tracking)**.** *Let the inner state obey*

$$w_{t+1} = w_t + a_t\{h(w_t; \mathcal{S}_t) + M_{t+1}\}. \tag{35}$$

Assume deterministic positive step sizes with $\sum_t a_t = \infty$ and $\sum_t a_t^2 < \infty$; adapted iterates; predictable modes $\mathcal{S}_t$; and martingale-difference noise with uniformly bounded conditional second moment. Suppose all iterates lie in a compact set D, every mode has a locally Lipschitz mean field, and D is forward invariant for its ODE. Assume finitely many modes, each with a Lyapunov-stable equilibrium $w^\star(\mathcal{S})$ attracting every point of D. Finally, either mode changes stop, or their successive spacings on the clock $s_t = \sum_{u<t} a_u$ tend to infinity. For every $\zeta > 0$, almost surely,

$$\frac{1}{s_T} \sum_{t<T} a_t \mathbf{1}\{\|w_t - w^\star(\mathcal{S}_t)\| > \zeta\} \longrightarrow 0. \tag{36}$$

Proof. The martingale $\sum_{t<n} a_t M_{t+1}$ has bounded total second moment and converges almost surely. Its increments over all sufficiently late finite clock windows therefore vanish uniformly. Boundedness of h on D, $a_t \to 0$, and the local Lipschitz property give the corresponding Euler approximation error. Applying the integral form of the ODE and Gronwall's inequality shows that, on any fixed-length interval without switching, the interpolation of the recursion uniformly approaches the ODE started at the same point. The constants can be chosen uniformly over the finite set of modes.

Fix ζ. Stability provides a smaller neighborhood of each equilibrium whose ODE trajectories stay inside the $\zeta/2$ neighborhood. Attraction and compactness give a common finite time H after which every trajectory starting in D is inside an even smaller neighborhood. To see uniformity, choose an entry time for each initial point, use continuity of the finite-time flow to obtain a neighborhood with the same entry property, and take a finite subcover of D. Stability then preserves the entry property at later times. Taking the maximum over modes gives the common H.

On sufficiently late unswitched windows, the approximation error is smaller than both neighborhood margins. After the first window of length H, the stochastic interpolation is inside the inner neighborhood. Applying the same argument on consecutive windows keeps their endpoints inside it and their intervening paths within distance ζ. Thus each switch contributes at most a fixed transient clock length, up to vanishing discretization errors. If $N(s)$ counts switches before clock time s, diverging spacings imply $N(s)/s \to 0$: eventually every complete spacing exceeds any prescribed constant. The total transient fraction is consequently bounded by a vanishing initial fraction plus $O(HN(s)/s)$, proving Eq. (36). If switching stops, the argument gives convergence to the final equilibrium. □

We hold the running statistics, such as the ledger and the reward normalization, fixed within each mode, so that each mode has an autonomous mean field. The finite-round accounting identity of Lemma A.5 holds without these conditions.

A.7 Randomized contrasts, bootstrap means, and decisions

Lemma A.7 (Randomization)**.** *For a fixed task, policy, and candidate, assign exactly n_v of G rollout slots to the visible context uniformly at random, independently of their potential outcomes. Assume*

no interference between slots. Then

$$\begin{aligned}\mathbb{E}[z_x \mid x,k] &= p^+ - p^- = \Delta_k(x),\\ \mathbb{E}[\bar{Y} \mid x,k] &= p^- + \tfrac{n_v}{G}\Delta_k(x).\end{aligned} \tag{37}$$

For fixed slot-level potential outcomes, the design variance is

$$\mathrm{Var}_{\mathrm{assign}}(z_x) = \frac{S_1^2}{n_v} + \frac{S_0^2}{n_h} - \frac{S_\Delta^2}{G}, \tag{38}$$

where the finite-population variances use denominator $G-1$. Under independent superpopulation rollout sampling, the unconditional variance given the task is $V_+/n_v + V_-/n_h$.

Proof. Let I_i mark a visible assignment and let $Y_i(1), Y_i(0)$ be fixed potential outcomes. Since $\mathbb{E}I_i = n_v/G$, the two observed part means have expectations equal to their corresponding population means. Their difference is the population mean treatment effect. Averaging over potential-outcome sampling yields the first identity, and the weighted average of the part means yields the second.

For the variance, $\mathrm{Var}(I_i) = n_v n_h/G^2$ and $\mathrm{Cov}(I_i, I_j) = -n_v n_h/[G^2(G-1)]$ for distinct slots. Write $z_x = \sum_i I_i c_i - \sum_i Y_i(0)/n_h$ with $c_i = Y_i(1)/n_v + Y_i(0)/n_h$. Substitution of these covariances gives $n_v n_h \sum_i (c_i - \bar{c})^2/[G(G-1)]$. Expanding this expression and using $S_\Delta^2 = S_1^2 + S_0^2 - 2S_{10}$ proves Eq. (38). Under independent rollout sampling, the two observed part means concern disjoint independent slots, so their variances add to $V_+/n_v + V_-/n_h$. □

A re-test group adds a withdrawn skill k to the visible part, so by the same lemma its contrast estimates $p_{K_x\cup\{k\}}(x) - p_{K_x}(x)$. The design is blocked by task: the contrast removes the baseline term at a fixed task, while task selection can still change the average effect. A fixed mean group success does not identify $\Delta_k(x)$, since different p^- values offset different effects in Eq. (37). For example, write $q_v = n_v/G$ and choose any $c \in (0,1)$. The probabilities $p^- = c - q_v d$ and $p^+ = c + (1-q_v)d$ give mean group success c and effect d. For all sufficiently small positive d, both are valid probabilities. Taking $d = 0$ gives the same mean and zero effect. Thus the nonidentifiability is exact even under a fixed group ratio, while the randomized contrast distinguishes these cases in expectation.

What adaptive records estimate. Predictable task and skill selection does not destroy within-group randomization. To see what it preserves, index the successive records for a fixed skill version by j and let $\mathcal{H}_{j-1}$ contain the information available before the next tested group is generated. Define $\mu_j = \mathbb{E}[z_j \mid \mathcal{H}_{j-1}]$, including the current task-selection law in that expectation. Then $e_j = z_j - \mu_j$ is a martingale difference, and for deterministic n,

$$\mathbb{E}\left[\left(\bar{z}_n - \frac{1}{n}\sum_{j=1}^{n}\mu_j\right)^2\right] \le \frac{1}{n}. \tag{39}$$

For $i < j$, conditioning on $\mathcal{H}_{j-1}$ gives $\mathbb{E}[e_i e_j] = 0$. Also $z_j \in [-1,1]$ implies $\mathbb{E}[e_j^2 \mid \mathcal{H}_{j-1}] = \mathrm{Var}(z_j \mid \mathcal{H}_{j-1}) \le 1$. Expanding the square proves the bound, so the ledger mean tracks the average of the predictable effects μ_j.

For a current target μ_{now}, the error is this sampling term plus the drift $n^{-1}\sum_j \mu_j - \mu_{\mathrm{now}}$; more records shrink the former, and the 30-cycle evidence window of Appendix C keeps old records from dominating the latter.

Bootstrap mean. Treat the ledger as an indexed collection of n contrast records $z_1, \ldots, z_n$, allowing repeated task occurrences. For weights $w \sim \mathrm{Dirichlet}(1,\ldots,1)$, define $\Delta^{\mathrm{BB}} = \sum_j w_j z_j$. Conditional on the records,

$$\begin{aligned}\mathbb{E}_{\mathrm{BB}}\Delta^{\mathrm{BB}} &= \bar{z},\\ \mathrm{Var}_{\mathrm{BB}}(\Delta^{\mathrm{BB}}) &= \frac{\sum_j (z_j - \bar{z})^2}{n(n+1)}.\end{aligned} \tag{40}$$

Indeed, $\mathbb{E}w_j = 1/n$, $\mathrm{Var}(w_j) = (n-1)/[n^2(n+1)]$, and distinct weights have covariance $-1/[n^2(n+1)]$; expanding the weighted sum proves both identities. Under a stationary exchangeable record model, this is the Bayesian-bootstrap posterior of the mean observed-contrast distribution. That mean equals the mean latent skill effect when randomization is unbiased on the same scope. No latent-effect likelihood is needed to define this mean.

For the adaptive ledger, we use the same Dirichlet construction as the posterior of the recorded comparison model, and the decisions below minimize posterior risk under it.

Lemma A.8 (Bayes decisions)**.** *Fix a posterior Π over a scope-specific mean effect Δ. With losses Δ^+ for withdrawal and ϵ for retention, withdrawal minimizes posterior expected loss exactly when $\mathbb{E}_\Pi[\Delta^+] \le \epsilon$, allowing withdrawal at a tie. With*

loss $1-\alpha$ *for admitting a nonbeneficial skill and* α *for withholding a beneficial one, admission is optimal exactly when* $\Pi(\Delta \le 0) \le \alpha$*, ties included.*

Proof. The withdrawal and retention risks are precisely $\mathbb{E}_\Pi[\Delta^+]$ and ϵ. For admission, put $q = \Pi(\Delta \le 0)$. The two risks are $(1-\alpha)q$ and $\alpha(1-q)$; their comparison reduces to $q \le \alpha$. Finally $\mathbb{E}_\Pi \Delta \le \mathbb{E}_\Pi \Delta^+$, proving the signed-mean posterior bound used in Eq. (33). □

The positive part is taken *after* forming the mean effect, and convexity places the resulting risk between two simpler quantities, $\bar{z}^+ \le \mathbb{E}_{\mathrm{BB}}[(\Delta^{\mathrm{BB}})^+] \le n^{-1}\sum_j z_j^+$. The rule thus bounds the modeled aggregate harm over the test scope. Minimum record counts restrict eligibility, the withdrawal-first precedence resolves overlap between admission and withdrawal, and trial expiration (Appendix C) is a separate management rule.

Versioning a skill by identifier, version, and content hash keeps evidence from pooling across edits to that skill, and the 30-cycle evidence window keeps it close to the current Director, context, and curriculum; the drift d_k of Eq. (33) carries the rest into the round cost. The prior pseudo-record enters only the exploration distribution Π_k^0 of Thompson selection, so decisions use the data-only Π_k, and the minimum-record gate keeps an untested skill from qualifying for a decision.

A.8 The anchored curriculum and distinct pairs

Lemma A.9 (Anchored curriculum)**.** *Fix a finite candidate list, a full-support reference* ρ_0*, finite conditional expected rewards* $\bar{R}_x$*, and* $\tau > 0$*. Over all distributions on the list, the unique maximizer of* $\sum_x \rho_x \bar{R}_x - \tau \mathrm{KL}(\rho\|\rho_0)$ *is the Gibbs distribution in Eq. (10). For two independent picks* $a, b \sim \rho_\phi$*, with conditional reward means depending only on the respective task,*

$$\mathbb{E}\left[\tfrac{1}{2}(R_a - R_b)(s_a - s_b)\right] = \nabla_\phi \sum_x \rho_\phi(x)\bar{R}_x, \tag{41}$$

where $s_x = \nabla_\phi \log \rho_\phi(x)$ *and rewards are held fixed when differentiating.*

Proof. With $Z = \sum_x \rho_0(x) e^{\bar{R}_x/\tau}$, the objective is $\tau \log Z - \tau \mathrm{KL}(\rho\|\rho^\star)$, proving uniqueness. Expanding the pair estimator, the two self-terms each have expectation $\nabla_\phi \sum_x \rho_\phi(x)\bar{R}_x$. The cross-terms vanish by independence and $\mathbb{E}[s_x] = 0$, and the factor one half gives Eq. (41). □

For the stated distinct-pair sampler, put $p_x = \rho_\phi(x)$ and assume at least two candidates with positive probability. Keeping the first pick and redrawing the second from the remaining candidates gives

$$\Pr(a, b) = \frac{p_a p_b}{1 - p_a}, \qquad a \ne b. \tag{42}$$

The unordered-pair probability sums both orders. Under the same task-local reward-mean assumption, the expectation of the raw estimator is

$$\begin{aligned} &\sum_{a<b} p_a p_b \, w_{ab} (\bar{R}_a - \bar{R}_b)(s_a - s_b), \\ &w_{ab} = \tfrac{1}{2}\left(\frac{1}{1-p_a} + \frac{1}{1-p_b}\right). \end{aligned} \tag{43}$$

This follows directly by multiplying the unordered-pair probability by the symmetric half-difference estimator. The independent-pick gradient is the same sum with $w_{ab} = 1$. These pair-dependent factors cannot generally be absorbed into a single learning rate. For pair-dependent rewards, replace $\bar{R}_a - \bar{R}_b$ by $\mathbb{E}[R_a - R_b \mid \{a, b\}]$, with reward labels attached to tasks rather than pick order.

A two-candidate example already separates the stationary points. Let $\bar{R}_1 - \bar{R}_2 = 1$, $\rho_0 = (1/2, 1/2)$, $\tau = 1$, and parameterize p_1 by its logit. The raw distinct-pair reward gradient is $1/2$, whereas the independent-pick gradient is $p_1(1-p_1)$. At the Gibbs logit of one, the KL gradient equals $p_1(1-p_1)$, so the raw distinct-pair regularized gradient remains nonzero. Gibbs optimality thus characterizes the ideal objective, while the stationary point of the practical update also depends on the pair weights, reward scaling, and trust region.

If both selected tasks supply training groups, the pooled task marginal within a list is

$$\nu(x) = \tfrac{1}{2} p_x \left(1 + \sum_{a \ne x} \frac{p_a}{1 - p_a}\right), \tag{44}$$

obtained by averaging first- and second-position marginals. Conditioning further on selection of skill k gives its test-task scope. Finally, a smaller expected reward implies a smaller Gibbs ratio $\rho^\star(x)/\rho_0(x)$, and the absolute probability also scales with the reference probability $\rho_0(x)$.

A.9 The three propositions

Proof of Proposition 1. Lemma A.5 proves the accounting statement for every finite sequence. Appending a test-time step $\mathcal{S}_{T+1} = \varnothing$ with $\theta_{T+1} = \theta_T$ makes its left side the zero-skill success under ρ

for any run, with the benefit of the remaining scaffold as the last outer cost. Lemma A.6 proves the tracking statement under its assumptions, and Lemma A.4 identifies the context mixture as q_x, or q'_x under separate last-skill centering. □

Proof of Proposition 2. Apply Lemma A.1 to the policy on K^- to obtain gain $\beta p^-(1-p^-)+O(\beta^2)$, or $\sqrt{2\delta p^-(1-p^-)}+O(\delta)$ at a common small budget, which grows with ℓ_{rec} to first order; for a parametric Director, Eq. (21) bounds the accessible part. Lemma A.3 identifies $4p^-(1-p^-)$ with normalized expected energy. Lemma A.2 gives the shared-tilt optimum of the quadratic allocation problem. With stochastic Workers, Eq. (20) instead governs graph-controllable progress. The operational shrinkage, transfer, and collapse terms turn this target into the expected reward of Lemma A.9. □

Proof of Proposition 3. The first identity of Lemma A.7 proves conditional unbiasedness. Averaging it over the actual test-task distribution proves unbiasedness for that distribution's mean effect. The second identity mixes the post-withdrawal baseline with the skill effect, as in Eq. (12), while the contrast cancels the baseline within the task. Bootstrap aggregation and Lemma A.8 then give the decisions of Eq. (14), and Eq. (33) carries them to the round costs. □

B Estimators and Implementation Details

Shrinkage and compute factor. The success rate $\tilde{p}_x$ in Eq. (9) shrinks toward the running hidden-part success rate of the task's dataset with strength m, an exponential moving average over previous cycles with decay 0.9 that starts at one half. The positive term of Eq. (9) is multiplied by $(\bar{T}_d/T_q)^\eta$, where T_q is the mean duration of the task's rollouts and $\bar{T}_d$ the running mean duration of the dataset, so tasks are compared within a dataset by learning progress per unit of compute. A task without both parts, for instance one for which no skill is retrieved, is scored by ℓ of the shrunk success rate of the whole group, with the same collapse penalty.

Prior record and cost term. The Thompson draw of Eq. (13) adds one pseudo-record from a wide zero-mean prior to the bootstrap, so a skill with no data is drawn from the prior alone. The Director's reward subtracts λ times the Workers' normalized cost from correctness; the contrasts z_x always use raw correctness.

Director update. The Director's loss is the clipped surrogate of PPO on the advantages of Eq. (5), whitened over the batch, which rescales every task by the same factor, and averaged over tokens, plus a KL anchor to the base model. Samples whose advantage is below a small threshold in magnitude, mostly from groups whose rollouts agree in correctness and differ only in cost, are dropped except for a small share kept for the anchor.

Proposer update. Each candidate list yields two distinct picks a and b: draw a from the choice policy and draw b from that policy conditioned on $b \neq a$. The advantage of a is half the reward difference $R_P(a)-R_P(b)$, scaled to unit root mean square over the batch, and b takes its negative; the loss adds τ times the KL of the list's choice distribution to the reference. A trust region drops the policy term once the pair's tilt $(\log\rho_\phi(a)-\log\rho_\phi(b))-(\log\rho_0(a)-\log\rho_0(b))$ passes a bound in the direction the advantage pushes, so the choice within a list stays stochastic.

Graph canvas. Agents, layers, and relations are edited through the action set of Definition 2, and the canvas validates each edit before it is applied.

C Training Loop and Skill Ledger

Algorithm 1 lists one round of R^{2}Orch; every step cites the equation it implements. An LLM editor revises skills, the least helpful first, and distills new ones from groups in which both parts fail, while withdrawals follow Eq. (14). A skill that the editor adds or revises enters the scaffold on trial under a new ledger key (identifier, version, and content hash), so its records start afresh; records for an unchanged skill identity remain pooled across Director updates. $\mathcal{N}_t$ admits it, a trial skill still unadmitted after $3n_{\min}$ records is discarded, and $\mathcal{W}_t$ is checked first, so the two rules never act on one skill in the same round. A task with $K_x = \varnothing$ runs its G rollouts without skills and adds no record. Each posterior uses the records of the last 30 cycles, and 30% of the tasks re-test withdrawn skills, which $\mathcal{N}_t$ can restore.

D Experimental Details

Benchmarks and splits. The six IID benchmarks, HotpotQA (Yang et al., 2018), NQ-Open (Kwiatkowski et al., 2019; Lee et al., 2019), MedQA (Jin et al., 2021), AIME, MBPP+ (Austin

Algorithm 1 One round of R^{2}Orch

Require: Director π_θ, Proposer ρ_ϕ, base models π_{ref}, ρ_0, scaffold $\mathcal{S}_t$ with summary s_t, ledger $\mathcal{D}_t$

for each candidate list C of the round **do**
  draw two distinct tasks from $\rho_\phi(\cdot \mid C, s_t)$ ▷ Appendix B
  for each selected task x **do**
    retrieve $K_x \subseteq \mathcal{S}_t$; on a re-test task, add one withdrawn skill to K_x as k_x ▷ Appendix C
    otherwise, if $K_x \neq \varnothing$, draw k_x by Thompson sampling; draw $\mathcal{H}_x$ uniformly, or set $\mathcal{H}_x = \varnothing$ if $K_x = \varnothing$ ▷ Eq. (13)
    set $K_i \leftarrow K_x$ for all i; for $i \in \mathcal{H}_x$, set $K_i \leftarrow K_x \setminus \{k_x\}$
    run the Director under (x, K_i) and execute its graph on the Workers; record correctness y_i, $i \le G$
    compute $R_P(x)$ using Eq. (9) or the skill-free fallback
    if $K_x \neq \varnothing$ **then**
      $z_x \leftarrow \bar{y}_{\mathcal{V}} - \bar{y}_{\mathcal{H}}$; append (k_x, z_x) to $\mathcal{D}_t$ ▷ Eq. (11)
$\theta \leftarrow$ clipped step on the advantages of Eq. (5) with the KL anchor to π_{ref}
$\phi \leftarrow$ distinct-pair step of Appendix B with the KL anchor to ρ_0 and the trust region on the tilt
for each skill k in $\mathcal{S}_t$ or under re-test with $n_k \ge n_{\min}$ **do**
  $\Pi_k \leftarrow$ Eq. (13); withdraw k if $k \in \mathcal{W}_t$, otherwise admit or restore it if $k \in \mathcal{N}_t$ ▷ Eq. (14)
discard each trial skill still unadmitted after $3n_{\min}$ records
if a revision candidate with a defined posterior exists, revise the one with the lowest posterior benefit
distill new candidates from groups in which both parts fail
return $\theta, \phi, \mathcal{S}_{t+1}, \mathcal{D}_{t+1}$

Symbol	Meaning	Value
G (n_h, n_v)	rollouts per task (hidden, visible)	5 (3, 2)
ϵ	withdrawal tolerance on the success rate	0.02
α	admission risk	0.05
$n_{\min}$	minimum records per decision	12
B	Bayesian-bootstrap draws	2000
–	Thompson prior standard deviation	0.25
–	evidence window of the posterior (cycles)	30
–	share of tasks that re-test withdrawn skills	0.3
γ	discount when only the visible part succeeds	0.5
λ_P	penalty when all rollouts agree	0.2
m	shrinkage strength	2
η	compute-factor exponent	0.5
λ	Worker-cost weight in the Director reward	0.005
τ	Proposer KL anchor	1.0
–	trust region on the Proposer's tilt (nats)	4.0
–	Director KL anchor; PPO clip (low, high)	0.005; (0.2, 0.28)
–	learning rate (Director, Proposer), cosine decay	10^{-5}, 10^{-5}
–	base model (Director, Proposer, Workers)	Qwen3.5-9B
–	LoRA rank	64
–	tasks per cycle	24 (2 lists of 8 per benchmark, 2 picks each)
–	Director updates	200
–	sampling temperature (Director, Proposer, Worker)	1.0, 1.0, 0.3

Table 4: Hyperparameters of R^{2}Orch.

et al., 2021; Liu et al., 2023), and ALFWorld (Shridhar et al., 2020), supply the training tasks and the IID tests; AIME trains on the pooled problems of earlier years and is tested on AIME 2026. The six OOD benchmarks, TriviaQA (Joshi et al., 2017), MuSiQue (Trivedi et al., 2022), GPQA (Rein et al., 2023), MATH-Hard (Hendrycks et al., 2021), SWE-Bench (Jimenez et al., 2024), and WebShop (Yao et al., 2022), are used only for testing. Each test set has 128 items (AIME 2026: 30) disjoint from the training tasks, so one run's 0/1 metric lies on a $k/128$ grid ($k/30$ for AIME 2026) and a five-run mean on a $k/640$ grid ($k/150$). The training pools and sizes are released with the code. The training curves in Figure 4a use a separate held-out set of 32 tasks from each IID dataset (AIME: 30 problems of earlier years), disjoint from the training tasks and distinct from the IID and OOD test sets.

Metrics. HotpotQA, NQ-Open, TriviaQA, and MuSiQue report exact match (EM) and F1 of the final answer, with Ans EM in Table 2; MedQA, AIME 2026, GPQA, and MATH-Hard report accuracy; MBPP+ reports Pass@1; ALFWorld and WebShop report the success rate (SR); and SWE-Bench reports the share of resolved issues.

Baselines and fairness. Qwen3.5-9B (Qwen Team, 2026) and DeepSeek-V4-Flash are prompted directly. SFT and GRPO (Shao et al., 2024), the latter with dynamic sampling, train Qwen3.5-9B

on the same training tasks, and each keeps the configuration whose trained weights score best under its default recipe. The other baselines are workflow search (AFlow (Zhang et al., 2025)), RL-trained orchestration (AgentFlow, FlowSteer (Li et al., 2026; Zhang et al., 2026c)), skill internalization (SKILL0, PATS (Lu et al., 2026b; Shi et al., 2026)), and self-play with skill evolution (SESA, MetaSkill (Fu et al., 2026; Wang et al., 2026)), and each runs the best configuration reported in its paper. All methods share the frozen Workers, training tasks, and compute budget; R^{2}Orch is tested at zero skills.

Ablation variants. Each ablation changes one component and keeps the rest of R^{2}Orch fixed, including the Workers, the training tasks, the training rollouts, and the zero-skill test. In each group of Table 2, the first row removes the whole component: GRPO only for Recursive Self-Play Orchestration, a failure reward for Recursive Learnability, and a fixed scaffold for Skill Withdrawal. The other rows remove one part of the orchestration (the skills or the Proposer's picks), replace only the Proposer's reward, or replace only the withdrawal rule, whose thresholds are tuned on validation.

Proposer objectives. Figure 5a–e replace only the Proposer's objective, by uniform sampling, dynamic sampling (DAPO (Yu et al., 2026)), selective rollout (GRESO (Zheng et al., 2026)), learnability on the current scaffold (SFL (Rutherford et al., 2024)), a solve-rate reward (Absolute Zero, AZR (Zhao et al., 2026)), or an uncertainty reward (R-Zero (Huang et al., 2026)), and keep the Director update, the scaffold, the withdrawal rule, and the training rollouts fixed; the Uniform, DAPO, SFL, and R^{2}Orch rows of Figure 5a repeat Table 2. Every objective trains 24 groups of five rollouts per cycle, DAPO's discarded rollouts count toward its cost, and scores are five-run means at zero skills.

Backbone transfer. For Figure 4e, the Director keeps the weights trained with Qwen3.5-9B Workers, and only the Workers are replaced by each of six other backbones, with skills off. Frozen prompts the backbone directly, and +R^{2}Orch runs the same backbone under the trained Director.

Training dynamics. Figure 4a evaluates the checkpoint of every training cycle on the held-out set, once with the current skills and once with none. Figure 4b counts the distinct graph types the Director builds over the six IID datasets. A candidate task is unseen if it has never been attempted; otherwise its previous success rate p over its earlier rollouts makes it unsolved ($p = 0$), learnable ($0 < p < 1$), or solved ($p = 1$). Figure 4c sums the Proposer's probability over the candidates of each class, and Figure 4d gives its probability on solved candidates as a moving average over 10 cycles, beside the untrained model with the same prompt and random choice (the share of solved candidates).

Selection evidence and withdrawals. Each candidate list of eight tasks yields two distinct picks (Appendix B). Figure 5e counts the 2,220 picks of seen tasks over cycles 50–199 by their previous success rate, and a new group is mixed if its rollouts contain at least one success and one failure. These counts include both picks of each list, the second drawn from the remaining candidates (Eq. (42)), whereas Figure 4d gives the policy's probability on solved candidates as a moving average over 10 cycles. Figure 5f measures the held-out gain of a dataset's three original seed cards, used together, over no cards, once at checkpoint 0 and as the mean over the 10 checkpoints around each withdrawal with its 95% interval. Each row names the withdrawal cycle and the card; *restored* marks the MedQA topology card, which re-testing restored after its withdrawal at cycle 6. This is dataset-level evidence; each withdrawal follows the posterior of the withdrawn card's contrast (Eq. (14)).

Hyperparameters. Table 4 lists every constant that appears as a symbol in the main text; the PPO clip and the learning rate set the step size β.

Compute. R^{2}Orch trains on A100 GPUs for 200 cycles, about 106 A100 hours measured in training, twice its 100-cycle cost. Figure 5c,d report the selection cost of the Proposer objectives: extra and total rollouts per cycle, rollouts and tokens per effective group, and A100 hours per 100 cycles. An effective group mixes successes and failures, so its GRPO advantages are not all zero.

Repeated runs. Table 1 reports the mean and standard deviation over five runs, and Table 2 and Figures 4e and 5a report five-run means.